The Goldbach–Chen Frontier: A Singular-Series Cancellation behind a Finite-Range Half-Power

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June 2026

Abstract

The binary Goldbach conjecture asks whether every even $N > 2$ is a sum of two primes; Chen's theorem allows the second summand to be a prime or a semiprime. Writing $R_1(N)$ and $R_2(N)$ for the number of representations $N = p+q$ with q prime and with q a semiprime, a log–log regression of R_2/R_1 against the Hardy–Littlewood singular series $\mathfrak{S}(N)$ assigns the semiprime channel an apparent exponent $\beta_2 \approx \frac{1}{2}$, strikingly unlike the $\beta_1 = 1$ of the prime channel. We show this half-power is an artifact of finite range, not of structure. Decomposing a semiprime $q = rs$ by its smaller prime factor, each channel inherits the same singular series $\mathfrak{S}(N)$ that governs R_1 (an elementary Euler-product identity), so $\mathfrak{S}(N)$ enters R_2 with exponent 1—exactly as in R_1—and cancels in the ratio of main terms, leaving a pure Mertens channel sum $W_f(N) \sim \log\log N$ with $R_2/R_1 = W_f(N)(1 + o(1))$. The contribution is the object, not the local computation: forming the channel ratio R_2/R_1 at fixed N, in which $\mathfrak{S}(N)$ cancels exactly, in contrast to the shifted-correlation literature, which fixes a gap and averages over the summand, where the two singular series do not cancel. This single cancellation organizes the whole picture—the channel ratio, the sign reversal of the effective exponent across arithmetic-complexity layers, the residual collapse, and the Chen-rescue threshold. Computing R_1, R_2 to 10^9 by a block-convolution sieve shows the effective exponent is not fixed: it drifts upward as $\beta_2(N) = 1 - c(N)/\langle R_2/R_1\rangle$ with deficit $\sim 1/\log\log N$, a drift a moving-block bootstrap makes statistically decisive and a singular-series-matched Cramér surrogate reproduces.

We are explicit about status. Unconditional are (i) the cancellation identity, and (ii) a linear almost-all asymptotic $R_2(N) = \mathrm{main}(N)(1 + o(1))$ for all but $o(X)$ even N, from a Vaughan–Parseval minor-arc second moment together with the equidistribution of semiprimes in arithmetic progressions (Siegel–Walfisz). Upgrading this to the log-regression statement $\beta_2(N) \to 1$ requires one further input, isolated as an explicit, unproven lower-tail hypothesis on R_2/R_1; the functional form $1 - \beta_2 \asymp 1/\log\log N$ is proved in the Cramér model and, for the primes, conditionally on a quantitative Hardy–Littlewood input. The goal is not a proof of Goldbach but the cancellation identity, a reproducible artifact it explains, and an unconditional almost-all asymptotic for the semiprime channel; cross-disciplinary reframings are recorded in a supplement.

2020 Mathematics Subject Classification. 11P32 (primary); 11N36, 11A41, 11Y35, 11-04 (secondary).

Keywords. Goldbach–Chen frontier; Hardy–Littlewood singular series; semiprimes; circle method; experimental number theory; almost-all asymptotics.

1 Introduction

The binary Goldbach conjecture states that every even integer $N > 2$ can be written as the sum of two primes. The conjecture remains open. However, several nearby facts are known. Helfgott

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proved the ternary, or weak, Goldbach conjecture: every odd integer greater than 5 is the sum of three primes [5]. Oliveira e Silva, Herzog and Pardi verified the binary Goldbach conjecture computationally up to $4 \cdot 10^{18}$ [10]. Chen's theorem proves that every sufficiently large even integer can be written as

$$N = p + q,$$

where p is prime and q is either prime or the product of at most two primes [1]; the semiprime channel and the "E_2" numbers it counts are classical sieve objects [3], and for the broader additive-bases context we refer to Nathanson [9]. Recent sieve-theoretic work also studies intermediate statements between Chen's theorem and the binary Goldbach conjecture [7].

This project focuses on the transition between the Goldbach representation

$$N = p + \text{prime}$$

and the Chen representation

$$N = p + \text{semiprime}.$$

The guiding question is not merely whether a representation exists, but how the available representations are distributed, how often the second term is genuinely prime, how often it is semiprime, and whether a representation can be selected in a smooth way as N varies.

Main idea. For each even N, define the number of representations with q prime and with q semiprime. Then study the empirical object

$$N \longmapsto \{(p, q) : p + q = N,\ p \in \mathbb{P},\ q \in \mathbb{P} \cup \mathcal{S}_2\}.$$

This is a discrete object, so it is not continuous in the ordinary analytic sense. Nevertheless, one can ask whether the induced empirical measures converge weakly and whether normalized counts have stable asymptotics. This gives a precise mathematical language for the intuition that the representation may "jump" between prime and semiprime behavior.

Summary of findings. Sections 4–4.5 report the results of implementing this program; all numbers are reproducible from the accompanying code. The main outcomes are:

1. **One identity behind the picture (§2.1, §4.2).** The organizing fact is an elementary, unconditional Euler-product identity: the per-channel singular series factorizes (Lemma 1), so $\mathfrak{S}(N)$ *cancels exactly* in the main-term ratio R_2/R_1, leaving the Mertens channel sum $W_f(N) \sim \log\log N$ with $R_2/R_1 = W_f(N)(1 + o(1))$. This one cancellation governs the channel ratio, the layer sign reversal (§4.3), the residual collapse, and the Chen-rescue threshold. Its most visible consequence resolves a finite-range puzzle. Counts are computed for all $N \leq 2 \cdot 10^6$ by FFT and locally in windows up to 10^9 by a block-convolution sieve: a finite-range log–log fit assigns the semiprime channel an apparent exponent $\beta_2 \approx \frac{1}{2}$, unlike the $\beta_1 = 1$ of the prime channel, yet scaling to 10^9 shows this is *not* structural ($\mathfrak{S}(N)$ enters R_2 with exponent 1). The effective exponent drifts to 1 as $\beta_2(N) = 1 - c/\langle R_2/R_1\rangle_N$ with an $\mathcal{O}(1/\log\log N)$ deficit (validation $(1-\beta_2)\langle R_2/R_1\rangle \approx 1.09$ to 1%; block bootstrap decisive); the deficit is multi-channel—deleting the heaviest channel ($3 \mid N$) removes only $\sim 10\%$ (§4.2), refuting a single-atom reading. What we record is this cancellation of $\mathfrak{S}(N)$ in the ratio, which the existence-oriented sieve and shifted-correlation literature has had no reason to isolate (Remark 2).
2. **Layers and sign reversal (§4.3).** For $\Omega(q) = k$, the effective exponent β_k falls through $1, \frac{1}{2}, \ldots$ and becomes negative for $k \geq 3$: $\mathfrak{S}(N)$ enhances low-complexity and suppresses high-complexity channels.

3. **Diagnostics (§4.4, §4.5).** Goldbach fragility is rare and always Chen-rescued; the semiprime factor geometry favours unbalanced semiprimes with a small prime factor; and the residual Goldbach–Chen balance exponent $a_{\mathrm{Res}}(N)$ correlates positively with $\mathfrak{S}(N)$. (Further location-law and valley diagnostics are in the supplement.)
4. **An unconditional almost-all asymptotic (Appendix B).** The cancellation identity (Lemma 1) is unconditional. A circle-method second-moment estimate—Vaughan's classical *prime* minor-arc bound plus the trivial L^2 size of the semiprime sum, no sieve (Prop. 2)—together with the major-arc evaluation (Lemma 6, the equidistribution of semiprimes in progressions, proved from Siegel–Walfisz) gives the *linear* asymptotic $R_2(N) = \mathrm{main}(N)(1+o(1))$ for all but $o(X)$ even N, *unconditionally* (Theorem 2). Upgrading to the log-regression statement $\beta_2(N) \to 1$ needs one further input (an explicit lower-tail hypothesis on R_2/R_1, beyond L^2); we state that gap rather than claim past it.
5. **A confirming experiment (§5).** The empirical circle method—the comet power spectrum as an empirical shadow of the major arcs—earns a place in the body by yielding a *new computation*; a smaller observation records a digit/Kummer carry signature of primality. The frontier admits a few further cross-disciplinary recastings (thermodynamic ground state, adversarial network, control problem, information channel, discrete surface); we are explicit that these are *one* finding—$\mathfrak{S}(N)$ sets the local densities—seen in several mirrors, not independent results, and we collect them, with the one structural negative, in the accompanying supplement rather than the body.

Status of the results: proven, conditional, conjectural. Because the central object is an empirical exponent that is easy to mistake for a structural one, we state once, explicitly, what is proved unconditionally, what is conditional and on which input, and what is left as a precise conjecture (Table 1); the body labels each result accordingly (**Theorem/Proposition** for proved statements, **Numerical Observation** for empirical ones, **Hypothesis/Conjecture** for what we do not prove).

2 Definitions and the singular-series identity

Let $\mathbb{P}$ denote the set of primes. For an integer $n \geq 2$, let $\Omega(n)$ be the number of prime factors of n counted with multiplicity. Thus $\Omega(12) = 3$ because $12 = 2^2 \cdot 3$.

Definition 1 (Semiprimes and Chen numbers)**.** *Define*

$$\mathcal{S}_2 := \{n \geq 2 : \Omega(n) = 2\}$$

to be the set of semiprimes, allowing squares of primes. Define also

$$\mathcal{P}_2 := \{n \geq 2 : \Omega(n) \leq 2\} = \mathbb{P} \cup \mathcal{S}_2.$$

Thus $\mathcal{P}_2$ is the set of integers with at most two prime factors counted with multiplicity.

For an even integer N, define the prime–prime representation count

$$R_1(N) := \sum_{p<N} \mathbf{1}_{\mathbb{P}}(p)\mathbf{1}_{\mathbb{P}}(N-p). \tag{1}$$

This counts ordered representations by choosing the first summand p. If unordered counts are desired, one may restrict to $p \leq N/2$.

Define the prime–semiprime representation count

$$R_2(N) := \sum_{p<N} \mathbf{1}_{\mathbb{P}}(p)\mathbf{1}_{\mathcal{S}_2}(N-p). \tag{2}$$

Statement	Status
Channel singular-series factorization $\mathfrak{S}_2(r,N) = 2C_2\mathfrak{S}(N)\frac{r-1}{r-2}$ (Lemma 1)	*Proved*, unconditional, elementary Euler-product computation
Main-term cancellation $R_{2,\mathrm{main}}/R_{1,\mathrm{main}} = W_f(N)$	*Conditional* on the stated Hardy–Littlewood / major-arc main-term input
Finite-range effective exponent $\beta_2 \approx \frac{1}{2}$ and its drift; fit (8)	*Numerical observation* (not extrapolated to a limit)
Functional form $1-\beta_2 \asymp 1/\log\log N$ (Theorem 3)	*Conditional* (quantitative Hardy–Littlewood, Assumption 1); the order is robust, the limiting constant open
Non-circular validation $(1-\beta_2)\langle R_2/R_1\rangle \approx \mathrm{const}$ (Obs. 1)	*Numerical observation* (control)
Linear almost-all $R_2 = \mathrm{main}(1+o(1))$ (Theorem 2)	*Proved*, unconditionally (Vaughan minor arc + semiprimes in progressions, Lemma 6); a standard circle-method consequence (Remark 6)
$\beta_2 \to 1$ for almost all N (Corollary 1)	*Conditional* on the lower-tail hypothesis
Lower-tail hypothesis	*Conjecture*, stated precisely (Conjecture 1)
Diagnostics of §4.4 (fragility, factor balance) and the circle-method experiment of §5 (spectrum, carry signature)	*Numerical observations*
Model statements in Appendix A (blocked channel, weak uniformity, least-unblocked prime, carry excess)	*Proved in the stated model*

Table 1: What is proved, what is conditional, and what is conjectured. The half-power $\beta_2 \approx \frac{1}{2}$ is a numerical observation; the structural exponent is 1 (Lemma 1).

Finally, define the Chen count

$$R_{\le 2}(N) := R_1(N) + R_2(N) = \sum_{p<N} \mathbf{1}_{\mathbb{P}}(p)\mathbf{1}_{\mathcal{P}_2}(N-p). \tag{3}$$

Remark 1. *The binary Goldbach conjecture is equivalent to $R_1(N) > 0$ for every even $N > 2$. Chen's theorem implies $R_{\le 2}(N) > 0$ for all sufficiently large even N.*

2.1 The singular-series identity

The Hardy–Littlewood singular series of N is

$$\mathfrak{S}(N) = \prod_{p \mid N,\, p>2} \frac{p-1}{p-2},$$

which governs the local density of Goldbach representations: the conjectural main term is $R_{1,\mathrm{main}}(N) = 2C_2\,\mathfrak{S}(N)\,I_1(N)$, with $C_2 = \prod_{p>2}\left(1-(p-1)^{-2}\right)$ the twin-prime constant and $I_1(N)$ the archimedean factor. The organising fact of this paper is that the *semiprime* channel inherits the *same* $\mathfrak{S}(N)$. Decompose a semiprime second summand by its smaller prime factor, $q = rs$ with $r \le s$ prime; each r defines a *channel*, a Goldbach-type count for the linear form $(s,\, N-rs)$ with its own two-form singular series.

Lemma 1 (Singular series of a channel)**.** *Let r be an odd prime and N even. The singular series of the pair of linear forms $s \mapsto (s,\, N-rs)$ is*

$$\mathfrak{S}_2(r,N) = \prod_{\ell} \frac{1-\nu_r(\ell)/\ell}{(1-1/\ell)^2}, \qquad \nu_r(\ell) = \#\{s \bmod \ell :\ s(N-rs) \equiv 0\},$$

and equals

$$\mathfrak{S}_2(r,N) = \begin{cases} 2C_2\,\mathfrak{S}(N)\,\dfrac{r-1}{r-2}, & r \nmid N,\\ 0, & r \mid N. \end{cases}$$

The proof is an elementary local computation (Appendix A). Two features matter: every *unblocked* channel ($r \nmid N$) carries the same $\mathfrak{S}(N)$ as R_1—not a smaller quantity, and not a fractional power of it; and the channels with $r \mid N$ are *identically blocked* ($\mathfrak{S}_2 = 0$), since $p = N - rs \equiv 0 \pmod r$ would force $p = r$. Summing the unblocked channel main terms and dividing by $R_{1,\mathrm{main}}$, the singular series *cancels exactly*:

$$\frac{R_{2,\mathrm{main}}(N)}{R_{1,\mathrm{main}}(N)} = W_f(N) := \sum_{\substack{3\le r\le\sqrt{N}\\ r\nmid N}} \frac{r-1}{r-2}\,w_r(N), \qquad w_r(N) = \frac{I_r(N)}{I_1(N)}, \tag{4}$$

with the exact archimedean channel weight $w_r = I_r/I_1$ (the ratio of the channel integral to the Goldbach one), for which the cancellation is exact and which is what Appendix B establishes; the elementary approximation $w_r \approx \frac{1}{r}\frac{\log N}{\log(N/r)}$ suffices for back-of-envelope estimates. This is a pure Mertens channel sum with $W_f(N) \sim \log\log N$. Under the major-arc input established unconditionally in Appendix B, the identity passes to the actual counts, $R_2(N)/R_1(N) = W_f(N)(1+o(1))$. This is the explanatory backbone of the paper: $\mathfrak{S}(N)$ enters R_2 with exponent 1, exactly as in R_1, so the "half power" $\beta_2 \approx \frac{1}{2}$ that a finite-range regression reports (§4.2) is an effective exponent inferred from the blocked channels, *not* a structural one. What we record is the *use* of this cancellation; the local computation itself is classical (Remark 2).

Symbol	Meaning
$\mathbb{P},\ \mathcal{S}_2,\ \Omega(n)$	primes; semiprimes ($\Omega = 2$); number of prime factors with multiplicity
$R_1(N),\ R_2(N)$	ordered counts of $N = p + q$ with q prime, resp. semiprime (1)–(2)
$R_{\le 2} = R_1 + R_2$	Chen count
$\mathfrak{S}(N) = \prod_{p\mid N, p>2} \frac{p-1}{p-2}$	Hardy–Littlewood singular series; C_2 the twin-prime constant
$\mathfrak{S}_2(r,N)$	channel singular series of $(s, N - rs)$ (Lemma 1)
$I_r(N),\ I_1(N)$	archimedean (channel / Goldbach) integrals; $w_r = I_r/I_1$
$W_f(N) = \sum_{r\nmid N} \frac{r-1}{r-2} w_r$	blocked Mertens channel sum, $\sim \log\log N$ (4)
$\beta_2,\ \beta_k$	effective singular exponents (Table 3)
$\theta(N),\ C(N)$	prime share $R_1/R_{\le 2}$; Chen surplus $R_2/(R_1+1)$

Table 2: Notation.

3 The Goldbach–Chen frontier and its classical heuristic

The empirical object is the split $R_{\le 2}(N) = R_1(N) + R_2(N)$ into genuinely Goldbach and genuinely semiprime representations. We track a few statistics. The *Chen surplus* $C(N) = R_2(N)/(R_1(N)+1)$ measures how far semiprime representations outnumber prime ones; the *prime share* $\theta(N) = R_1(N)/(R_1(N)+R_2(N))$ is the complementary ratio; and, for a threshold k, the *k-fragile* integers $\mathcal{F}_k(X) = \{4 \le N \le X \text{ even} : R_1(N) \le k,\ R_2(N) > 0\}$ are those for which Goldbach is sparse while Chen still rescues. (Here R_1 is the *ordered* count (1), so the counts $|\mathcal{F}_k|$ are convention-dependent—an unordered R_1, or dropping the $R_2 > 0$ condition, changes them; with this convention $|\mathcal{F}_1| = 1$, the integer being $N = 6$.) For a semiprime $q = ab$ with $a \le b$ prime, the *factor balance* $B(q) = \log a/\log q \in (0, \frac{1}{2}]$ is near 0 when q has a small prime factor and near $\frac{1}{2}$ when q is balanced.

The classical Hardy–Littlewood heuristic [4] fixes the normalisation we use as a control, not as a new derivation: the Goldbach main term is

$$R_1(N) \approx 2C_2\,\mathfrak{S}(N)\,\frac{N}{\log^2 N}, \tag{5}$$

with C_2 the twin-prime constant and $\mathfrak{S}(N)$ as in §2.1. For the semiprime channel, Landau's count $\#\{n \le x : \Omega(n) = 2\} \sim x \log\log x/\log x$ [6] gives $R_2(N) \asymp K_2(N)\,N\log\log N/\log^2 N$ with a local correction K_2: at a first heuristic level semiprime representations outnumber prime–prime ones by a factor $\sim \log\log N$, so the share $\theta(N)$ should drift slowly downward. Section 4.2 makes the response of the ratio R_2/R_1 to $\mathfrak{S}(N)$ precise and resolves the apparent half-power. The location law of q/N and the type-labeled measures—whether the normalized representation clouds stabilize weakly to a near-uniform law—are confirmed in the supplement; here we go straight to the singular-series response.

4 Empirical results and the singular-series regularity

This section reports the outcome of implementing the program above. All numbers are reproducible from the accompanying code (see the Data and code availability statement; the drivers `run_counts.py`, `run_scaling.py`, `run_betalaw.py`, `run_estimator_audit.py`); unless stated otherwise they use the range $X = 2\cdot 10^6$.

4.1 Computational setup and validation

The counts R_1 and R_2 are obtained for *all* $N \le X$ at once as additive convolutions,

$$R_1 = \mathbf{1}_{\mathbb{P}} * \mathbf{1}_{\mathbb{P}}, \qquad R_2 = \mathbf{1}_{\mathbb{P}} * \mathbf{1}_{\mathcal{S}_2},$$

evaluated by FFT in $O(X\log X)$. For $X = 2\cdot 10^6$ this takes about 15 seconds including sieves, diagnostics and figures. The FFT counts agree exactly with a brute-force enumeration on random samples (the rounding error of the transform is far below 1/2), and Goldbach's condition $R_1(N) > 0$ is verified throughout the range.

Prior computational work, and what is new. Large-scale computation on the Goldbach side is well established: Oliveira e Silva, Herzog and Pardi verified $R_1(N) > 0$ to $4\cdot 10^{18}$ [10], and the Goldbach comet—the count $R_1(N)$ itself—has been tabulated for decades. Our contribution is on the *semiprime channel*. To our knowledge, the systematic computation of $R_2(N)$ (the Chen surplus), the prime share $\theta(N)$, and—centrally—the singular-series *response* of the ratio R_2/R_1 has not been carried out at this scale; the block-convolution sieve (below) takes these channel counts to $X = 10^9$ in $2\cdot 10^6$-windows. This is emphatically *not* a record for R_1 (it is far short of [10]); the point is the first systematic measurement of how the *semiprime* channel responds to $\mathfrak{S}(N)$, the object this paper is about.

4.2 The semiprime channel and the finite-range half-power

The Hardy–Littlewood prediction (5) attaches to R_1 the full singular series $\mathfrak{S}(N) = \prod_{p|N,\,p>2}(p-1)/(p-2)$. Regressing $\log\big(R_1/(N/\log^2 N)\big)$ against $\log\mathfrak{S}(N)$ recovers, empirically, a slope 1.000, and dividing by $\mathfrak{S}(N)$ collapses the coefficient of variation of the normalized count from 0.39 to 0.015 (a 25× reduction); this is a control that reproduces (5) and the band structure of the Goldbach comet.

The channel observation concerns R_2. We measure how strongly the relative semiprime share tracks $\mathfrak{S}(N)$ through the regression of $\log(R_2/R_1)$ on $\log\mathfrak{S}(N)$, which uses R_1 itself as a meter for $\mathfrak{S}(N)$ and is therefore independent of any chosen normalization.

Regression notation. To avoid confusing the raw regression slope with the effective singular exponent, we write

$$m_2(X) := \mathrm{slope}_{5000 \le N \le X}\left(\log \frac{R_2(N)}{R_1(N)} \quad \text{on} \quad \log \mathfrak{S}(N)\right), \qquad \beta_2(X) := 1 + m_2(X).$$

Thus $\beta_2 = 1$ means that R_2 carries the same singular-series power as R_1, while $m_2 = 0$ means that the ratio R_2/R_1 is uncorrelated with $\mathfrak{S}(N)$. We call β_2 the *effective singular exponent* of R_2: it is an effective exponent inferred from a regression slope, and the raw slope is $m_2 = \beta_2 - 1$. As Lemma 1 shows below, $\mathfrak{S}(N)$ enters R_2 with exponent 1 (exactly as in R_1) and *cancels* in the ratio R_2/R_1, so R_2/R_1 is not a power of $\mathfrak{S}$ and the value $\beta_2 \approx \frac{1}{2}$ reported in finite ranges is a finite-size correction, not a structural exponent (developed after Table 4). All tables below report β_2, not the raw slope m_2, using one estimator: ordinary least squares, cumulative over all even $5000 \le N \le X$. The OLS standard errors quoted are *descriptive only*—consecutive even integers are arithmetically dependent, so no inferential claim rests on them; the inferential basis is the moving-block bootstrap of §4.2 (Table 6), which respects that dependence. The variance appendix (§B.2) uses instead a *local* window centred at X, running ≈ 0.02–0.03 higher at the same X—the distinction is between an average over $[5000, X]$ and a measurement at the leading edge. Several exponents appear below (cumulative, local, first-moment, Cramér, and the layered β_k); they are all the *same* regression read on different data, and Table 3 collects them so they are not confused.

Estimator	Definition	Representative values
β_2 (cumulative)	1+ OLS slope of $\log(R_2/R_1)$ on $\log \mathfrak{S}$, even $N \in [5000, X]$	$0.45, 0.48, 0.50$ ($X = 5{\cdot}10^5, 10^6, 2{\cdot}10$
β_2 (local)	same slope over the window $N \in [X, X + 2{\cdot}10^6]$	$0.55 \to 0.61$ ($X = 6{\cdot}10^6 \to 10^9$; Tab.
$\beta_2^{(1)}$ (first moment)	from the conditional responses A_ℓ, B_ℓ (§4.2)	$0.53, 0.56, 0.59$ ($X = 2{\cdot}10^6, 2{\cdot}10^7, 10$
$\beta_2^{\text{Cramér}}$ (surrogate)	β_2 of the matched-density Poisson model	$0.53 \to 0.75$ ($X = 10^6 \to 10^{50}$; Fig. 9)
β_k (layers)	1+ slope of $\log(R_k/R_1)$ on $\log \mathfrak{S}$	$1.0, 0.5, -0.3, \ldots$ (Tab. 7)

Table 3: The several exponents, all the same regression read on different data or by a different functional. The cumulative β_2 is the headline; the local, first-moment, and Cramér versions are independent cross-checks of the same drift, and β_k is the layered generalization.

X	β_2	R^2_{OLS} of the regression
$5 \cdot 10^5$	0.452	0.91
$1 \cdot 10^6$	0.475	0.91
$2 \cdot 10^6$	0.497	0.92
$4 \cdot 10^6$	0.513	0.93
$8 \cdot 10^6$	0.529	0.93

Table 4: Effective singular exponent $\beta_2 = 1 + m_2$ of R_2, where m_2 is the cumulative OLS slope of $\log(R_2/R_1)$ on $\log \mathfrak{S}$ over $[5000, X]$; the table reports β_2, not the raw slope $m_2 = \beta_2 - 1$. The value sits near $\frac{1}{2}$ in this range and *rises* with X—already a warning that it is not a fixed exponent (contrast $\beta_1 = 1.000$, which is genuinely scale-invariant). The singular series explains about 92% of the variance of $\log(R_2/R_1)$. OLS standard errors are descriptive only (nearby N are arithmetically dependent).

Over this range, the best finite-range log–log normalization is consistent with the effective fit (near $X \sim 10^6$, the effective exponent $\frac{1}{2}$ drifting to 1; §4.2 below)

$$R_2(N) \;\propto\; \mathfrak{S}(N)^{1/2} \frac{N}{\log^2 N} W_0(N), \qquad W_0(N) = \sum_{\substack{3 \le r \le \sqrt{N} \\ r \text{ prime}}} \frac{r-1}{r-2} \frac{1}{r} \frac{\log N}{\log(N/r)}, \tag{6}$$

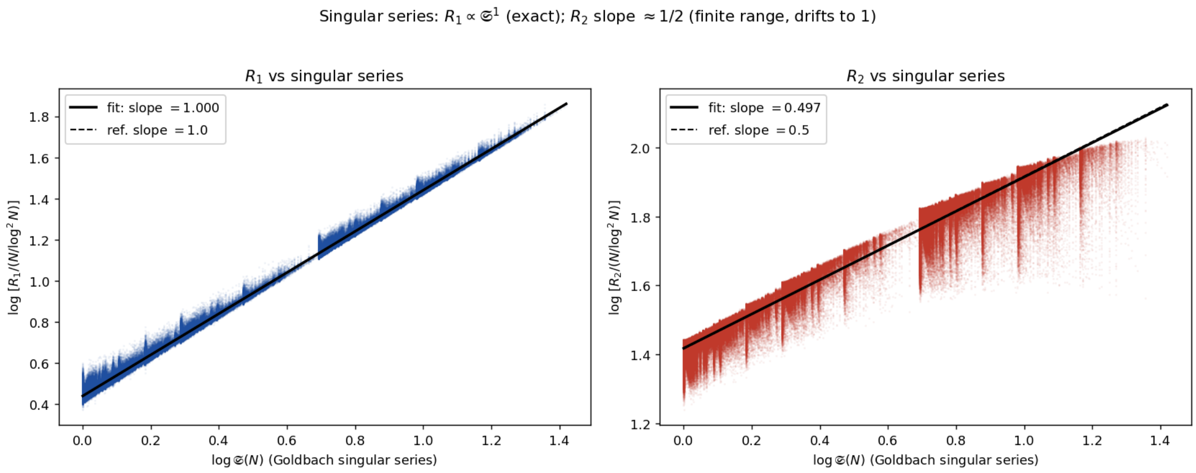

Figure 1: Left: $\log[R_1/(N/\log^2 N)]$ against $\log \mathfrak{S}(N)$ has slope 1.000 (Hardy–Littlewood). Right: the same regression for R_2 gives effective exponent $\beta_2 = 0.497$ at $X = 2 \cdot 10^6$, near the reference line $\beta = 1/2$—but this is the *finite-range* effective exponent of a non-power-law ratio (it drifts toward 1; Fig. 2), not a fixed power. The vertical striations are the discrete values of $\mathfrak{S}(N)$.

with $W_0(N) \sim \log\log N$ by Mertens' theorem [14]; here W_0 is the *unblocked* Mertens channel sum, ranging over all small primes r. The theoretically relevant object is instead the *blocked* channel sum W_f, which omits the channels $r \mid N$ (where $p = N - rs \equiv 0 \bmod r$ forces $p = r$),

$$W_f(N) = \sum_{\substack{3 \le r \le \sqrt{N} \\ r \in \mathbb{P},\ r \nmid N}} \frac{r-1}{r-2} \frac{1}{r} \frac{\log N}{\log(N/r)}, \tag{7}$$

the same sum W_f that governs the appendix derivation, where it is the channel-cancellation factor $R_2/R_1 = W_f(N)(1+o(1))$ (Theorem 3, conditional). A notational caveat: in (6), (7) and Table 5 the channel weight is the *elementary* back-of-envelope form $w_r \approx \frac{1}{r}\frac{\log N}{\log(N/r)}$ displayed above, whereas the appendix and Theorem 2 use the *exact* archimedean weight $w_r = I_r(N)/I_1(N)$ that it approximates; the elementary form overstates I_r/I_1 by a constant factor for small r (quantified numerically in §B.3), which shifts the constants but not the qualitative comparison. We use the elementary weight in Table 5 for comparability with the finite-range fit (6); recomputing the four dispersions with the exact weight I_r/I_1 on a subsample changes the constants only slightly ($0.184/0.167/0.019/0.0135 \to 0.184/0.167/0.019/0.016$) and leaves the ordering—and the blocked full-inheritance normalization as the tightest—unchanged. Table 5 compares the dispersion of R_2 under four normalizations.

Interpretation: a quantitative Chen rescue effect. Because the finite-range effective exponent satisfies $\beta_2 < 1$, over this range the relative semiprime share falls with the singular series, $R_2/R_1 \propto \mathfrak{S}(N)^{-(1-\beta_2)}$ (locally $\approx \mathfrak{S}(N)^{-1/2}$): where Goldbach is locally rich (large $\mathfrak{S}$), the prime channel dominates, whereas where Goldbach is *fragile* (small $\mathfrak{S}$) the semiprime channel is proportionally most abundant. This makes the Chen-rescue intuition—the semiprime channel matters most exactly where primes are scarce—quantitative as an *effective finite-window scaling*, with the deficit $1 - \beta_2 \asymp 1/\log\log N$ vanishing asymptotically (so β_2 is an effective exponent, not a fixed power; §4.2 opening). Heuristically the effective exponent sits below 1 because writing $q = rs$ couples the divisors of N to the factor r: when a small prime $\ell \mid N$, the channel $r = \ell$ is *blocked* (then $p = N - \ell s \equiv 0 \bmod \ell$ forces $p = \ell$), removing its term from the sum over channels precisely when $\mathfrak{S}(N)$ is large (Lemma 1). This explains the direction $\beta_2 < 1$; the precise value is not $1/2$, as the scaling analysis below shows.

normalization of R_2	CV ($X = 2 \cdot 10^6$)
$N \log\log N / \log^2 N$ (heuristic of §3)	0.184
$2C_2\, \mathfrak{S}(N)\, N\, W_0 / \log^2 N$ (full $\mathfrak{S}$, *unblocked* W_0)	0.167
$2C_2\, \mathfrak{S}(N)^{1/2}\, N\, W_0 / \log^2 N$ (half power, W_0)	0.019
$2C_2\, \mathfrak{S}(N)\, N\, W_f / \log^2 N$ (**full inheritance, blocked channels**)	**0.0135**

Table 5: Dispersion of R_2 under four normalizations. The finite-range half-power normalization is descriptive rather than structural: it fits tightly (CV 0.019) only because it pairs the unblocked Mertens sum W_0 with the *wrong* power $\mathfrak{S}^{1/2}$. The theoretically relevant full-inheritance normalization is the *blocked*-channel expression $2C_2\mathfrak{S}(N)W_f(N)N/\log^2 N$, which carries the *full* singular series ($\beta = 1$) and fits *tighter still* (CV 0.0135); forcing the full series with the *unblocked* W_0 instead fits worst (0.167). The apparent 1/2 power thus summarizes, over a finite range, the covariance between the blocked channels $r \mid N$ and $\mathfrak{S}(N)$; it is not a limiting singular-series exponent.

Scaling to 10^9: the effective exponent drifts up like $\log\log X$. Measuring β_2 *locally* in a window of $2 \cdot 10^6$ even integers centred at X (computed by a block-convolution sieve, so that X can reach 10^9 while only the window is held in memory), the effective exponent does *not* settle at 1/2: it rises monotonically (Fig. 2). Eight windows from $6 \cdot 10^6$ to 10^9 are fit to within 0.0013 by

$$\beta_2(X) \;\approx\; -0.094(7) + 0.234(3)\, \log\log X \qquad (R^2_{\mathrm{OLS}} = 0.9987,\ 8 \text{ windows}), \tag{8}$$

the parenthesised figures being the last-digit standard errors of the two-parameter fit. Thus the half-power $\beta_2 \approx \frac{1}{2}$ is only the *local* value near $X \sim 10^6$, and the effective exponent is not a constant. The fit reaches $\beta_2 \approx 0.61$ at 10^9 (measured). We deliberately *do not* read a numerical limit off (8): extrapolating a $\log\log X$ trend fitted over three decades to, say, $X \sim 10^{46}$ (where it would touch $\beta_2 = 1$) is well outside the data and is not evidence—the robust content is the *functional form* $1 - \beta_2 \asymp 1/\log\log X$ and its derivation below, not a forecast of where the effective exponent crosses a particular value. The regression quality stays high throughout ($R^2_{\mathrm{OLS}} \approx 0.997$, CV of the half-power normalization $\lesssim 0.04$).

Statistical significance of the drift (block bootstrap). The inferential claim is the *slope*, not its absolute value, so we attach error bars that respect the arithmetic dependence of consecutive N (an i.i.d. bootstrap would understate it). A moving-block bootstrap over each window—blocks of $L = 10^3$ consecutive even integers, $B = 10^3$ resamples (`run_bootstrap.py`)—determines $\beta_2(X)$ to four significant figures: the 95% interval has half-width below 10^{-4} at *every* window (Table 6), widening only to $\sim 3 \cdot 10^{-4}$ as L ranges over 250–2000, because each window holds $\sim 10^6$ integers. Refitting the $\log\log X$ trend on each resample gives a drift slope that is positive with a 95% interval of width $< 10^{-3}$, excluding 0 by thousands of standard errors: the upward drift is statistically certain, not an artifact of the fit. The measurement error in β_2 is therefore negligible; the only genuine uncertainty is the long extrapolation of the functional form, which we do not perform (§C.2). The non-circular product $(1 - \beta_2)\langle R_2/R_1\rangle$ decreases monotonically and smoothly from 1.111 to 1.074 across $[6 \cdot 10^6, 10^9]$, again with negligible within-window CI—confirming *to* 10^9 (the previous validation stopped at $2 \cdot 10^8$) the bounded, slowly varying behaviour on which the derivation below rests.

A Cramér null control. Is the drift a property of the local blocking structure, or of subtler arithmetic correlations among the actual primes? A matched-density surrogate decides. Replace the true counts by independent Poisson variables with the singular-series-matched local means $\mathbb{E}R_1(N) = 2C_2\mathfrak{S}(N)N/\log^2 N$ and $\mathbb{E}R_2(N) = \mathbb{E}R_1(N) \cdot W_f(N)$, where W_f is the *blocked* channel sum (4) (the $r \mid N$ exclusion is the only arithmetic input), and rerun the identical β_2 regression

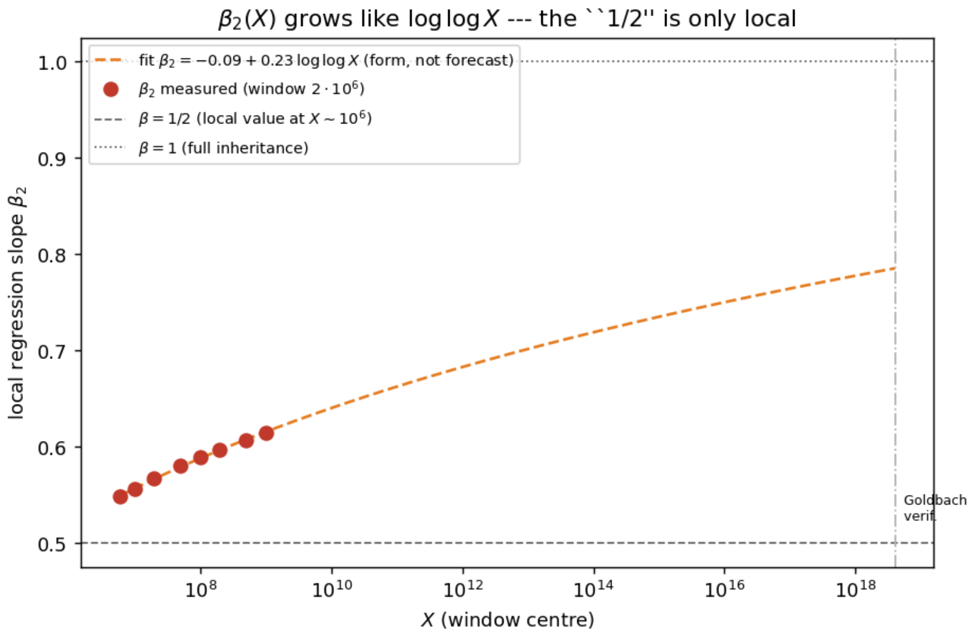

Figure 2: Local effective singular exponent $\beta_2(X) = 1 + m_2(X)$ for windows up to 10^9, with the $\log\log X$ fit (8) (the dashed extrapolation beyond 10^9 is shown only to display the functional form, not as a prediction). The effective exponent drifts steadily above 1/2 without sign of leveling off, consistent with $\beta_2 \to 1$.

(`run_nullcontrol.py`; no sieve to X is needed, as $\mathfrak{S}, W_f$ depend only on the small divisors of N). The surrogate reproduces the drift: $\beta_2^{\text{Cramér}}(X)$ rises $0.568 \to 0.624$ over $[6 \cdot 10^6, 10^9]$, tracking the measured $0.548 \to 0.614$ (Table 6) to within ≈ 0.02 and matching its slope. The drift is therefore a deterministic consequence of the local blocking encoded in W_f, not of higher-order correlations among the primes; the small residual offset is the elementary-weight/Jensen gap between W_f and the true channel counts.

An independent estimator: the first-moment exponent. The drift is also not an artifact of the log-regression estimator. A *first-moment* exponent $\beta_2^{(1)}$, built from the conditional responses $A_\ell = \langle R_1\rangle_{\ell|N}/\langle R_1\rangle_{\ell\nmid N}$ and $B_\ell = \langle R_2\rangle_{\ell|N}/\langle R_2\rangle_{\ell\nmid N}$ of the counts to each small prime ℓ (Appendix B; `run_firstmoment.py`)—a different functional of the data, not a regression slope—reproduces β_2 to a constant Jensen gap of ≈ 0.004: $\beta_2^{(1)} = 0.528, 0.563, 0.585$ at $X = 2 \cdot 10^6, 2 \cdot 10^7, 10^8$, versus the measured $\beta_2 = 0.531, 0.567, 0.589$. Two independently constructed estimators thus give the same drift.

A derivation: $\beta_2 \to 1$ with an $1/\log\log N$ deficit. The drift is not arbitrary. By the channel cancellation of §2.1 (Lemma 1), the main-term ratio is the Mertens channel sum $W_f(N)$ of (4), so $R_2(N)/R_1(N) = W_f(N)(1 + o(1))$ for the actual counts under the per-channel asymptotic. Hence the raw regression slope is $m_2 = \beta_2 - 1 = \operatorname{Cov}(\log W_f, \log\mathfrak{S})/\operatorname{Var}(\log\mathfrak{S})$, and the only coupling to $\mathfrak{S}(N)$ comes from the blocked channels $r \mid N$, each *removed* from W_f. Modelling $\mathbf{1}[\ell \mid N]$ as independent with probability $1/\ell$, $\operatorname{Var}(\log\mathfrak{S}) =: B$ is a convergent constant, while the removed terms contribute $\operatorname{Cov}(\log W_f, \log\mathfrak{S}) = -A/W_{f,0}$ with $A > 0$ constant and $W_{f,0} \approx \langle R_2/R_1\rangle \sim \log\log N$ the baseline sum (Theorem 3). Therefore

$$\beta_2(N) = 1 - \frac{c(N)}{\langle R_2/R_1\rangle_N} + o\left(\frac{1}{\log\log N}\right), \qquad c(N) = \tfrac{A}{B} = O(1)\ (\approx 1.1 \text{ in range; Fig. } 10), \quad (9)$$

X	$\beta_2(X)$	$(1-\beta_2)\,\langle R_2/R_1\rangle$
$6\cdot10^6$	0.5483	1.1105
$1\cdot10^7$	0.5564	1.1061
$2\cdot10^7$	0.5670	1.1002
$5\cdot10^7$	0.5800	1.0930
$1\cdot10^8$	0.5890	1.0882
$2\cdot10^8$	0.5974	1.0835
$5\cdot10^8$	0.6074	1.0780
$1\cdot10^9$	0.6143	1.0742

Table 6: Local effective exponent $\beta_2(X)$ (window of $2\cdot10^6$ even integers) and the non-circular product, to $X=10^9$. Each entry is a 95% moving-block bootstrap estimate (block length 10^3, $B=10^3$, `run_bootstrap.py`); the CI half-width is below 10^{-4} for β_2 and below $3\cdot10^{-4}$ for the product at every window, so the monotone upward drift of β_2 and the slow decrease of the product are both statistically decisive. The drift is the empirical core; the limiting constant (§C.2) is not.

so $\beta_2\to1$: R_2 *asymptotically inherits the full Goldbach singular series*, the approach being governed by $1/\log\log N$. This explains why finite ranges show $\beta_2\approx\frac{1}{2}$ and a slow apparent drift.

Numerical Observation 1 (Non-circular validation)**.** *The law* (9) *predicts that* $(1-\beta_2)\langle R_2/R_1\rangle$ *is constant. Measured over windows from* $6\cdot10^6$ *to* 10^9 *it equals* $1.11, 1.11, 1.10, 1.09, 1.09, 1.08, 1.08, 1.07$ *(Table 6), a coefficient of variation of* 1.1% *(versus* 2.2% *for* $(1-\beta_2)\log\log N$*): the correct "clock" is* $\langle R_2/R_1\rangle$*, exactly as* (9) *requires (Fig. 3). This is non-circular in the sense that no algebraic identity forces the product* $(1-\beta_2)\langle R_2/R_1\rangle$ *to be constant:* β_2 *is a covariance/variance slope and* $\langle R_2/R_1\rangle$ *a mean (both functionals of the same window), and their product holding to* 1% *is a property of the data, not a tautology.*

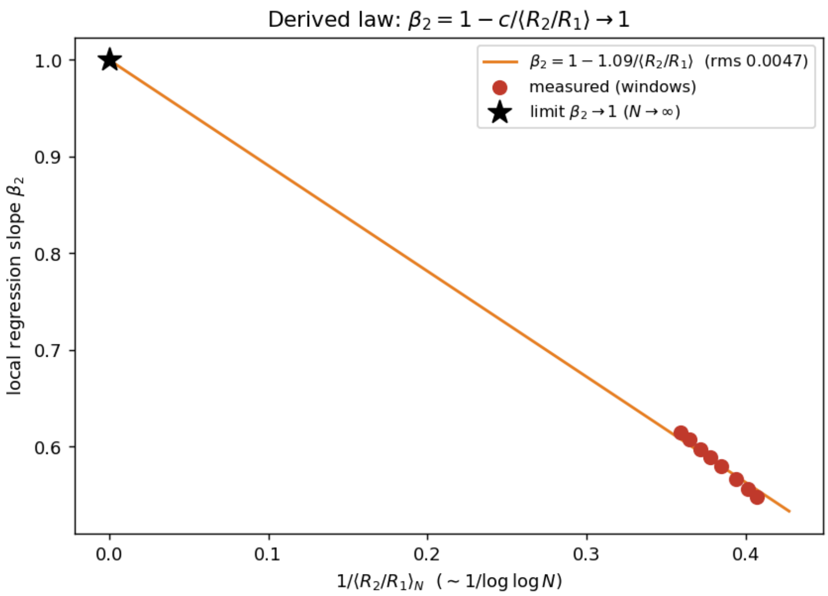

Figure 3: The derived law: β_2 against $1/\langle R_2/R_1\rangle\sim1/\log\log N$ is a straight line through the limit point $(0,1)$, i.e. $\beta_2\to1$.

Not a single-atom effect. A natural worry is that the whole deficit $1-\beta_2$ is the arithmetic of one prime: $3 \mid N$ simultaneously doubles $\mathfrak{S}(N)$ (the factor $\frac{3-1}{3-2}=2$) and deletes the heaviest semiprime channel $r=3$, so one might suspect $\beta_2 \approx \frac{1}{2}$ is just "what divisibility by 3 does" near $X \sim 10^6$. The data refute this. Recomputing β_2 on the subpopulation $3 \nmid N$ (the $r=3$ atom removed) gives, at $X = 2\cdot 10^6$, $\beta_2 = 0.547 \pm 0.0004$ against 0.495 on all N: the deficit $1-\beta_2$ falls by only about 10% (`run_estimator_audit.py`). The mechanism is real—conditional means confirm $\langle \log \mathfrak{S} \mid 3 \mid N\rangle - \langle \log \mathfrak{S} \mid 3 \nmid N\rangle = 0.69 \approx \log 2$ and a depressed R_2/R_1 on $3 \mid N$—but it accounts for a tenth of the deficit, not the bulk. The negativity of $\beta_2 - 1$ is genuinely a *sum over channels* $r = 3, 5, 7, \ldots$, each blocked on the sparse set $r \mid N$, exactly as $A(X) = \sum_r (\cdots)$ in (9); it is not the signature of a single residue class.

From $\beta_2 \to 1$ to a precise conjecture. The derivation and its validation show $1-\beta_2$ vanishing, but turning the *almost-all* $\beta_2(N) \to 1$ into a theorem needs one input beyond the unconditional second moment. We isolate it as a conjecture and state it in the four-part form a reader needs.

Conjecture 1 (Lower-tail hypothesis)**.** *For every $\delta > 0$, the density of even $N \in (X, 2X]$ with $R_2(N)/R_1(N) \le \delta\, W_f(N)$ tends to 0 as $\delta \downarrow 0$, uniformly in X.*

(i) Why it matters. This is precisely the gap between what we prove and the headline: by Corollary 1 it upgrades the *unconditional* linear asymptotic $R_2 = \mathrm{main}(1+o(1))$ (Theorem 2) to the log-regression statement $\beta_2(N) \to 1$. *(ii) Proved cases.* The linear almost-all asymptotic itself is unconditional (Theorem 2), and the L^2 second moment (Prop. 2) already bounds how often R_2 is *large*; only the one-sided lower tail—how often R_2/R_1 is anomalously *small*—is open. *(iii) Prior work, and why the natural attack fails.* The hypothesis is a lower-tail control beyond L^2. The natural route—a Chen-type lower bound on $R_{\le 2}$ minus the Montgomery–Vaughan value of R_1 [1, 8]—provably does *not* suffice: the unconditional Chen bound lies at the scale of R_1, a factor $\log\log N$ below the semiprime main term (§B.2), so subtracting R_1 is vacuous. What is needed is a lower bound on the *semiprime* channel at the correct $\log\log N$ scale, which the sieve does not currently supply. *(iv) What the data say.* The exceptional set is empirically *empty* at the 5% level up to $5\cdot 10^8$ (Table 9): in the computed range no even N has R_2/R_1 anywhere near the tail the conjecture forbids.

4.3 The singular exponent across arithmetic-complexity layers

The decomposition R_1, R_2 extends to layers $R_k(N) = \#\{p < N : \Omega(N-p) = k\}$, counting representations whose second summand has exactly k prime factors (Goldbach is $k=1$, Chen's semiprime channel is $k=2$). Measuring the singular exponent β_k of each layer – the slope of $\log(R_k/R_1)$ against $\log \mathfrak{S}(N)$, plus one – reveals a clean pattern (Table 7, Fig. 4):

k	1	2	3	4	5	6
share $\rho(k)$	15.5%	33.5%	28.7%	14.6%	5.5%	1.8%
β_k	1.00	0.50	−0.30	−1.38	−2.73	−4.04

Table 7: Layer profile and singular exponent at $X = 2\cdot 10^6$ (each regression has $R^2_{\mathrm{OLS}} \approx 0.92$). The exponent decreases monotonically and crosses zero between $k=2$ and $k=3$; it is *not* $1/k$.

Sign reversal of the singular response. The Goldbach singular series *enhances* the low-complexity channels ($\beta_1 = 1$, $\beta_2 \approx \frac{1}{2} > 0$) but *suppresses* the high-complexity ones ($\beta_k < 0$ for $k \ge 3$): where $\mathfrak{S}(N)$ is large, representations with a many-factored q become rarer. The mechanism is the same divisor effect seen for R_2, now run in reverse. When $\mathfrak{S}(N)$ is large, N is

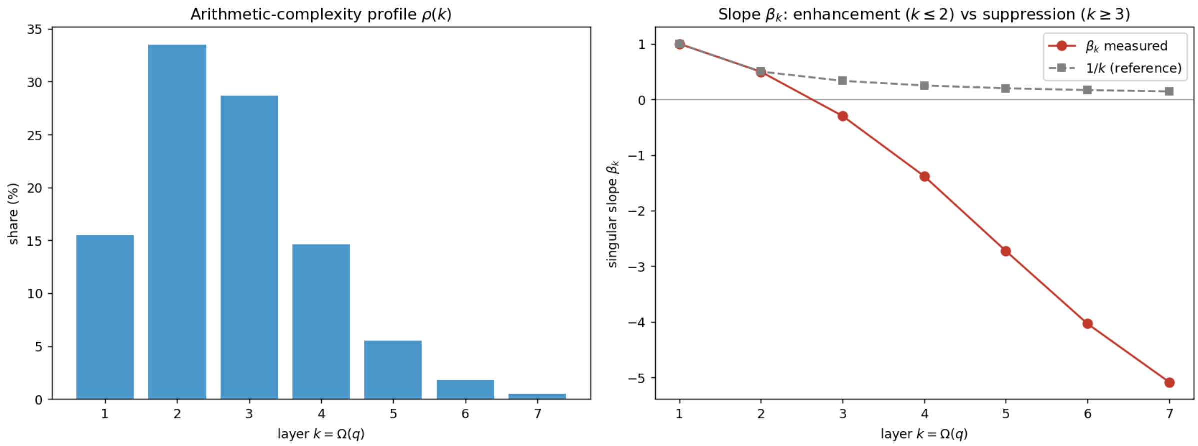

Figure 4: Left: the arithmetic-complexity profile $\rho(k)$ peaks at $k = 2$. Right: the singular exponent β_k decreases through $1, \frac{1}{2}, \ldots$ and becomes negative for $k \geq 3$, diverging from the naive $1/k$ reference.

divisible by many small primes; the admissible p avoid those residues, which forces $q = N - p$ to also avoid small prime factors, so q tends to have *fewer* prime factors. Thus a large singular series biases q toward low arithmetic complexity. All β_k drift upward with X at the same slow rate as β_2 (e.g. β_3 moves from -0.30 at $2 \cdot 10^6$ to -0.20 at $8 \cdot 10^6$), so the crossing point of $\beta_k = 0$ moves slowly to the right. We also find that the strongest "rescue" channel, $q =$ composite$\times$composite (both factors composite), is available for *every* even N in the range with on average $\sim 1.8 \cdot 10^4$ representations: the additive hierarchy is extremely redundant below the prime layer.

The generating-function mechanism. The whole pattern is one generating-function response. Encode the layers by $Z_N(z) = \sum_{p<N} \mathbf{1}_{\mathbb{P}}(p)\, z^{\Omega(N-p)}$, so that $[z^k]Z_N(z) = R_k(N)$ and $z = 1$ recovers the Chen count. In the blocked-channel model the log-density of the $\Omega = k$ layer is, to first order, linear in the channel fluctuation, and its singular-series response is $\beta_k(X) - 1 \approx -(k-1)\,\kappa_X$ with a single positive slope $\kappa_X = 1 - \beta_2$ (Prop. 7): positive below the typical factor count and negative above it, so β_k falls linearly through $1, \frac{1}{2}, \ldots$ and crosses zero near $k^\star = 1 + 1/\kappa_X \asymp \log\log X$. This is why a large $\mathfrak{S}(N)$ enhances the low-complexity channels and suppresses the high-complexity ones. The precise statement and the Ω-generating function are in Appendix C.4; no correlation-inequality machinery is needed—a Poisson linearization of the same blocked-channel model suffices.

4.4 Diagnostics

The prime share has mean $\bar{\theta} = 0.311$ and decreases slowly with N (the semiprime channel gains ground, as the $\log\log N$ factor predicts); the Chen surplus has mean $\overline{C} = 2.27$. Goldbach fragility is extremely rare: $|\mathcal{F}_1| = 1$, $|\mathcal{F}_2| = 3$, $|\mathcal{F}_3| = 5$, $|\mathcal{F}_5| = 14$, $|\mathcal{F}_{10}| = 49$ up to X; and *every* fragile integer observed has $R_2 > 0$, so Chen always rescues. The semiprime factor balance $B(q) = \log a / \log q$ has weighted mean 0.25, with 16% of the representation mass at $B < 0.1$ (a tiny prime factor) and 18% at $B > 0.4$ (near balanced): Chen representations are dominated by *unbalanced* semiprimes with a small prime factor (Fig. 5), answering the semiprime factor-geometry question of §3.

4.5 The residual Goldbach–Chen balance exponent

Inspired by the $(1 + a)$ interpolation of Li and Liu (every large even N is $p + rq$ with $r \leq q^{a-1}$, so that $(1+1)$ is Goldbach and $(1+2)$ is Chen), one can ask not whether a *fixed* exponent works,

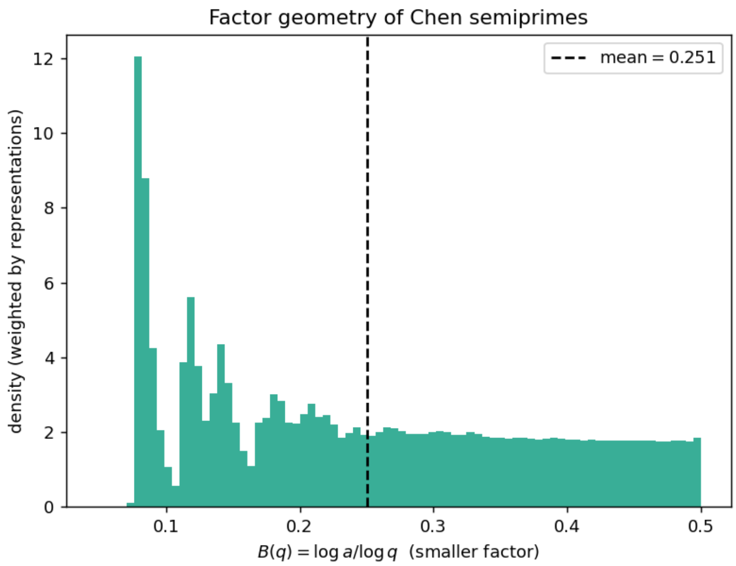

Figure 5: Representation-weighted distribution of the factor balance $B(q)$ of the semiprime second summand. Mass concentrates toward small B (small prime factor), not toward the balanced value 1/2.

but the smallest exponent *needed* for each N. For the nondegenerate Chen layer $N = p + rs$ (p, r, s prime, $r \leq s$), define the residual balance exponent and minimal multiplier

$$a_{\mathrm{Res}}(N) = 1 + \min_{N=p+rs} \frac{\log r}{\log s}, \qquad r_\star(N) = \min\{r \in \mathbb{P} : \exists\, p, s \in \mathbb{P},\ N = p + rs\}.$$

We compute both exactly for all even $N \leq 2 \cdot 10^6$ ($r_\star$ by FFT channel coverage; a_{Res} by a small-p scan, validated against full enumeration to 0 error). Findings (Fig. 6, $N \geq 1000$):

- **Typical residual collapse.** The median a_{Res} *decreases* with X ($1.115 \to 1.099 \to 1.088$ at $X = 10^5, 5 \cdot 10^5, 2 \cdot 10^6$), with $p_{99} = 1.182$ and a finite-range maximum of 1.385 – far below the Li–Liu threshold 1.9. The worst case is pinned by a few small "hard" N. In Li–Liu's terms this is a worst-case versus typical contrast: their theorem *guarantees* a representation with exponent $a \leq 1.9$ for every large N, whereas the *typical* N sits far inside that bound—the median Chen solution has $a \approx 1.30$ (supplement) and the *best* exponent per N is $a_{\mathrm{Res}} \approx 1.09$. For almost all N the $(1 + a)$ interpolation is therefore very close to Goldbach ($a = 1$), not to the proven upper bound.

- **The minimal multiplier is tiny.** $r_\star = 3$ for 68% of N and the dictionary $\{2, 3, 5, 7\}$ already covers 99.7%; $\{2, \ldots, 11\}$ covers 99.99%. This settles the multiplier-dictionary question empirically. Apart from the parity-degenerate $r = 2$ channel (which forces $p = 2$, a density-zero family), Appendix C.5 explains the odd multiplier dictionary through a least-unblocked-prime law: conditionally on fixed-channel almost-all asymptotics, the nondegenerate $r_\star(N)$ equals the least odd prime not dividing N for almost all even N (so $\Pr(r_\star \in \{3, 5, 7\}) \approx 0.99$ in the divisibility model).

- **Local obstruction.** $r_\star$ jumps when a small prime divides N: if $3 \mid N$ the channel $r = 3$ is blocked (then $p = N - 3s \equiv 0 \bmod 3$), and indeed the mean $r_\star$ is 4.59 for $3 \mid N$ versus 2.94 otherwise.

- **Link to the singular series.** a_{Res} correlates *positively* with $\log \mathfrak{S}(N)$ (correlation +0.59). The very divisibility of N by small primes that enlarges $\mathfrak{S}(N)$ (boosting Goldbach) also blocks the small-r Chen channels, so the residual balance is *worst exactly where Goldbach is locally richest*. This ties the balance exponent to the singular-series theme of §4.3. (The correlation +0.59 is a descriptive within-window statistic over consecutive even N, which are arithmetically dependent; it quantifies a direction and a rough strength, not an i.i.d. effect size. The analogous dependence for the central exponent β_2 is controlled by the block bootstrap of §4.2.)

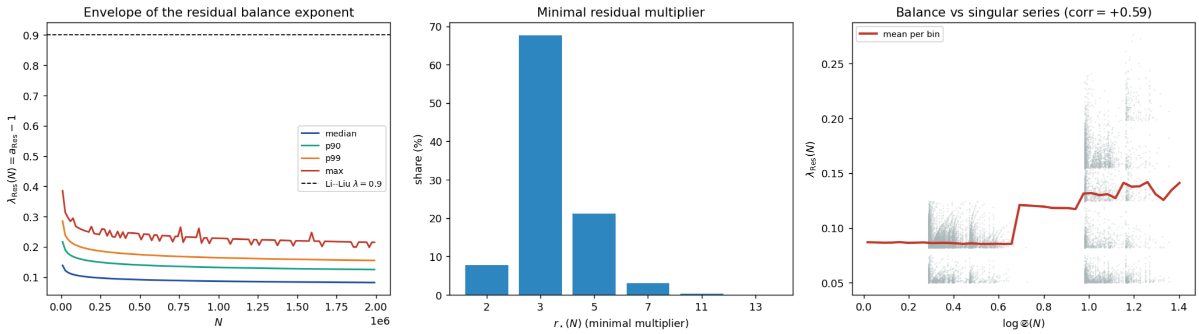

Figure 6: Left: percentile envelope of $\lambda_{\mathrm{Res}} = a_{\mathrm{Res}} - 1$ versus N, far below the Li–Liu line 0.9. Centre: the minimal multiplier $r_\star$ ($r = 3$ dominates). Right: λ_{Res} rises with the Goldbach singular series $\mathfrak{S}(N)$ (positive correlation).

The full geometry of the Chen surface $p + rs = N$ on which the exponent a_{Res} lives—a 2D cloud of solutions organised by the small prime factor r, with mass against the Goldbach edge—is sketched in the supplement.

5 A confirming experiment: the empirical circle method

One further experiment yields a genuinely *new* measurement and ties directly to the circle-method backbone: the power spectrum of the comet is an empirical shadow of the major arcs. (A second, smaller probe—a carry geometry—is recorded as an observation below; the many cross-disciplinary recastings of the frontier re-express the same singular-series governance and are noted, with the one structural negative, in the supplement.)

The circle method decomposes additive problems into major arcs (rationals a/q with small q, carrying the singular series) and minor arcs (the pseudorandom remainder). One sees this *empirically*: detrending R_1 over a window of consecutive even integers and taking its power spectrum (Fig. 7, left) reveals sharp lines exactly at the arithmetic frequencies—the dominant peak at $\frac{1}{3}$ (period 6, the modulus 3), then $\frac{1}{5}, \frac{2}{5}$ (modulus 5), with heights decaying across primes in step with the singular-series factors $(\ell-1)/(\ell-2)$. These are the major-arc frequencies made visible. Dually, predicting the detrended $\log R_1(N)$ from the residue $N \bmod \prod_{\ell \le B} \ell$ explains 84.6% of the variance at $B = 3$ ($N \bmod 6$ alone), 95.0% at $B = 5$, and saturates toward the singular-series ceiling $R^2_{\mathrm{OLS}} = 0.999$ (Fig. 7, right): the comet is, to first order, a modulo-6 phenomenon, the prime 3 supplying 85% of all explainable structure. (These are descriptive R^2_{OLS} values over an arithmetically dependent window—a partition of within-window variance by residue class—not i.i.d. fits.)

Numerical Observation 2 (A digital signature of primality)**.** *A second, smaller probe records a genuinely new statistic. Adding $p + q = N$ in base b produces* $\#\mathrm{carries}(p, N-p) = (S_b(p) + S_b(N-p) - S_b(N))/(b-1)$*, which in base 2 is $v_2\binom{N}{p}$ (Kummer). Goldbach pairs carry* more *than random pairs $a + (N-a) = N$: in base 2 the excess is +0.97 carries* (9.9 *versus* 8.9)*, positive in every base tested* ($b = 2, \ldots, 16$) *and largest in small bases (Fig. 8). The origin is local—a prime's last digit is coprime to b, biasing $S_b(p) \equiv p \pmod{b-1}$ away from carry-avoiding*

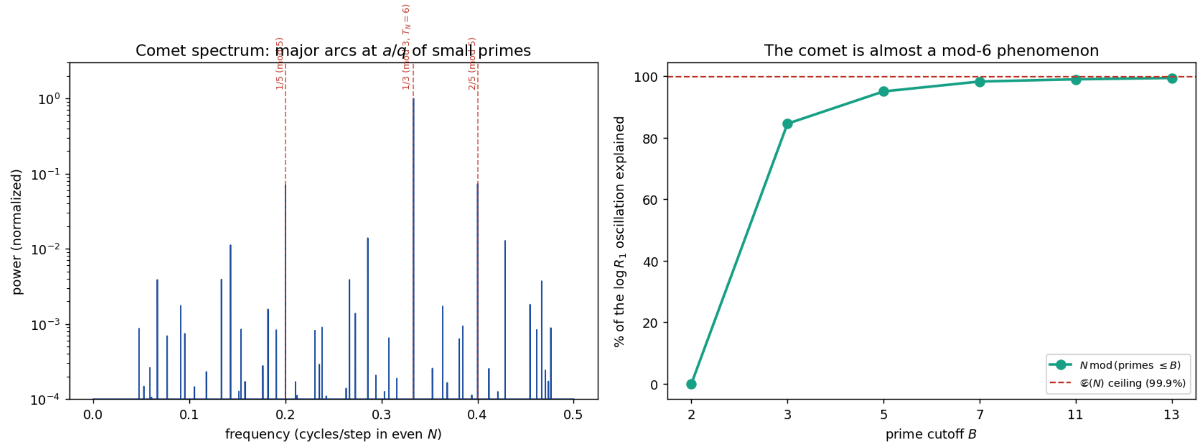

Figure 7: Empirical circle method. Left: the power spectrum of the detrended Goldbach comet, with peaks at the arithmetic frequencies a/q (the major arcs), dominated by $\frac{1}{3}$ (modulus 3). Right: the fraction of the comet's oscillation explained by N modulo the small primes, saturating at the singular-series ceiling; N mod 6 already gives 85%.

values—so the same constraints that produce the singular series leave a digital fingerprint; a local model in the supplement shows that restricting both terminal digits to reduced classes already raises the least-digit carry probability.

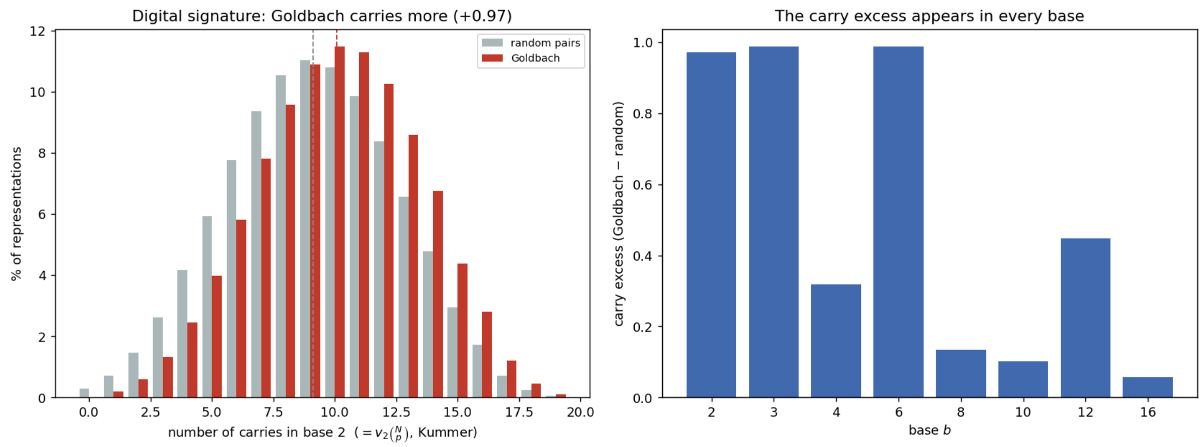

Figure 8: Carry geometry. Left: the distribution of binary carries $v_2\binom{N}{p}$ for Goldbach representations (red) versus random pairs (grey)—Goldbach is shifted to more carries (+0.97). Right: the carry excess is positive in every base.

6 Open problems and conjectures

We isolate, as precisely as the evidence allows, what an interested reader could try to settle.

1. **The lower-tail hypothesis** (Conjecture 1). Prove it—a one-sided lower-tail bound on R_2/R_1, beyond the unconditional L^2 control—and, by Corollary 1, the log-regression statement $\beta_2(N) \to 1$ follows. The four-part case (why it matters, what is proved, why the Chen−Montgomery–Vaughan attack provably fails, and the empirical support) is laid out at Conjecture 1; what is needed is a lower bound on the *semiprime* channel at the correct $\log\log N$ scale.

2. **The limiting constant** (§C.2). The order $1 - \beta_2 \asymp 1/\log\log X$ is robust, but the limiting constant of $(1-\beta_2)\langle R_2/R_1\rangle$ is correction-dominated and *not* determinable from data up to 10^9 (it is normalisation-dependent, ≈ 0.5–0.7). Is there a canonical normalisation with a clean limit?

3. **Layers** (§4.3). The Poisson model predicts $\beta_k(X) = 1 - (k-1)\kappa_X$ to leading order, with crossing $k^\star = 1 + 1/\kappa_X \asymp \log\log X$ (Prop. 7); this captures the slope near the top and the crossing layer but not the observed convex (accelerating) descent of the deeper β_k. Establish the second-order correction — i.e. the $O((k-1)^2\kappa_X^2)$ steepening — and prove the crossing drift unconditionally.

7 Conclusion

The organizing observation of this study is a single Euler-product identity and its consequence: decomposing the semiprime channel by its small prime factor shows that the Goldbach singular series $\mathfrak{S}(N)$ enters the main term of R_2 with exponent 1, exactly as it enters R_1 (Lemma 1), so $\mathfrak{S}(N)$ *cancels* in the main-term ratio and $R_2/R_1 = W_f(N)(1+o(1))$ under the stated major-arc input. The local factorization is an elementary computation; our contribution is to isolate the cancellation, write it down explicitly, and follow its consequences. The first is that the naturally read "$\beta_2 \approx \frac{1}{2}$" of a finite-range fit is not a structural exponent but a finite-range effective exponent inferred from the regression slope $m_2 = \beta_2 - 1$, drifting to 1 like $1 - c/\log\log N$; we verify this non-circularly, show it is a multi-channel effect (not the $3 \mid N$ atom), and reduce the limit to classical inputs—for the *linear* almost-all asymptotic $R_2 = \mathrm{main}(1+o(1))$ within the second-moment reduction developed in Appendix B, and, for the log-regression statement $\beta_2 \to 1$, modulo one explicitly stated lower-tail hypothesis (Corollary 1). A truncated-log proposition and a blocked-channel model (§C.3) sharpen this gap: the second moment already controls a stable logarithmic statistic and the model proves the $1/\log\log X$ deficit, while only the untruncated log-regression still requires lower-tail input.

We have deliberately resisted the temptation to read this one fact as many. The cross-disciplinary reframings are, except for the empirical circle method kept in the body (with a smaller carry-signature observation), the same singular-series governance re-photographed in different filters, and we have said so rather than count them as separate findings. The one structural negative—an empty additive-complexity topology—is reported as such.

This is not a proof of Goldbach. It is reproducible evidence, one cancellation identity made explicit, and an unconditional almost-all asymptotic for the semiprime channel, at the boundary between additive number theory and computational experimentation.

This appendix collects the analytic backbone in three parts: the unconditional singular-series identity and its moments (§A), the unconditional linear almost-all asymptotic for R_2 (§B), and the conditional program that upgrades it to $\beta_2 \to 1$ (§C). We cannot prove $\beta_2 \to 1$ unconditionally—the asymptotics of R_1, R_2 are themselves Hardy–Littlewood, and even $R_1(N) > 0$ is open—so the law is proved conditionally on a quantitative Hardy–Littlewood hypothesis (Theorem 3) and holds unconditionally in the Cramér random model.

A The singular-series identity

A.1 Notation and the statistic

Fix a window $\mathcal{N}_X = \{N \equiv 0\,(2) : X \le N < X + H\}$ with $H = H(X) \to \infty$, $H = o(X)$. Write $\mathbb{E}_X, \mathrm{Var}_X, \mathrm{Cov}_X$ for empirical moments over $\mathcal{N}_X$, $g_\ell = \log\frac{\ell-1}{\ell-2}$, and define the local singular exponent through the (least-squares) regression slope

$$\beta_2(X) - 1 := \frac{\operatorname{Cov}_X\big(\log(R_2/R_1),\, \log\mathfrak{S}\big)}{\operatorname{Var}_X\big(\log\mathfrak{S}\big)}. \tag{10}$$

A.2 Unconditional lemmas

Proof of Lemma 1 (§2.1). Recall $\mathfrak{S}_2(r,N) = \prod_\ell (1-\nu_r(\ell)/\ell)/(1-1/\ell)^2$ with $\nu_r(\ell) = \#\{s \bmod \ell : s(N-rs) \equiv 0\}$, and $C_2 = \prod_{p>2}(1-(p-1)^{-2})$. Compute $\nu_r(\ell)$. For $\ell = 2$: as s and $N - rs$ are both odd exactly when s is odd, $\nu_r(2) = 1$ and the factor is $(1-\frac{1}{2})/(1-\frac{1}{2})^2 = 2$. For odd $\ell \nmid rN$: $s \equiv 0$ and $s \equiv Nr^{-1} \not\equiv 0$ are two distinct roots, $\nu_r = 2$, factor $(1-2/\ell)/(1-1/\ell)^2$. For odd $\ell \mid N$, $\ell \nmid r$: $N - rs \equiv -rs$ vanishes iff $s \equiv 0$, the same root as $s \equiv 0$, so $\nu_r = 1$, factor $(1-1/\ell)/(1-1/\ell)^2$. For $\ell = r \nmid N$: $N - rs \equiv N \not\equiv 0$, so only $s \equiv 0$, $\nu_r = 1$, same factor. Taking $\prod_{\ell>2}(1-2/\ell)/(1-1/\ell)^2 = C_2$ as the base and converting each special factor relative to it multiplies by $(\ell-1)/(\ell-2)$; the special primes are exactly those dividing N (giving $\prod_{\ell\mid N} = \mathfrak{S}(N)$ when $r \nmid N$) together with $\ell = r$ (giving $(r-1)/(r-2)$). This proves the first case. If $r \mid N$ then $N - rs \equiv 0 \bmod r$ for *all* s, so $\nu_r(r) = r$, the factor $1 - r/r = 0$ vanishes, and $\mathfrak{S}_2(r,N) = 0$: the channel is identically blocked because $p = N - rs \equiv 0 \bmod r$ forces $p = r$. □

Remark 2 (Relation to standard sieve theory)**.** *Each ingredient of Lemma 1 is classical. The singular series of a system of linear forms is the Hardy–Littlewood computation [4] (see also [15]), and the decomposition of a semiprime by its prime factors is the staple device of the E_2 sieve [3]; for a single fixed channel $(s,\, N-rs)$ the evaluation of $\mathfrak{S}_2(r,N)$ is therefore elementary, and a reader familiar with the prime k-tuple heuristic will reconstruct Lemma 1 without difficulty. What we wish to record is not this local computation but its global consequence, which to our knowledge is not stated explicitly in the literature: that every unblocked channel inherits the* same *Goldbach singular series $\mathfrak{S}(N)$ that governs R_1—not a smaller quantity, and not a fractional power of it. The unconditional content is the singular-series identity itself: $\mathfrak{S}(N)$ cancels* exactly *between the Hardy–Littlewood main terms of R_2 and R_1, leaving the pure Mertens channel sum* (4)*; passing to the actual counts, $R_2(N)/R_1(N) = W_f(N)(1+o(1))$ requires the per-channel asymptotic and is conditional. That the singular-series cancellation seems not to have been written down is a matter of aim, not of difficulty. The sieve literature on Chen's theorem and on the $(1+a)$ interpolation [1, 3, 7] is directed at* existence*—lower bounds forcing $R_{\le 2}(N) > 0$, and upper bounds within a constant of the Hardy–Littlewood prediction. The conjectural prime–semiprime main term itself (the Hardy–Littlewood asymptotic for $R_{\le 2}$ and for R_2) is implicit in these heuristics, in the Chinese-school refinements of Chen's theorem (Pan and Pan [11], and the tradition of Cai and Lü), and in the prime/almost-prime correlation literature—the same singular-series machinery and the same local densities. Closest in spirit is the latter: Evans [2] proves Hardy–Littlewood asymptotics for the shifted correlation between primes n and almost-primes $n+h$ with $\Omega(n+h) = 2$, valid for almost all shifts h (by dispersion/Bombieri–Vinogradov machinery), whose singular series is the same Euler-product object as in Lemma 1. The crucial distinction is the* object*: a shifted correlation fixes a gap h and averages over n, so the two singular series do* not *cancel; our object fixes the sum $N = p + q$ and forms the channel* ratio *R_2/R_1, in which $\mathfrak{S}(N)$ cancels exactly—and that cancellation is the whole point. What none of the existence/lower-bound or shifted-correlation works isolates is this ratio and the exact cancellation of $\mathfrak{S}(N)$ in it: for the existence and upper-bound aims the singular-series content of R_2/R_1 is irrelevant and is never separated out. It becomes the natural object only once one asks how the relative semiprime share responds to $\mathfrak{S}(N)$, as the empirical regression of §4.2 forces. The identity is precisely what shows that the reproducible finite-range value $\beta_2 \approx \frac{1}{2}$ is not a structural half-power but an effective exponent inferred from the covariance between the blocked channels $r \mid N$ and $\mathfrak{S}(N)$ (Theorem 3): $\mathfrak{S}(N)$ enters R_2 with exponent* 1*, exactly as it does R_1. The new object is therefore the* ratio *R_2/R_1 and the exact cancellation of $\mathfrak{S}(N)$ in it. The local* *computation of $\mathfrak{S}_2(r, N)$ is, by itself, classical; what is new is isolating this cancellation and putting it to use—something the existence/upper-bound and shifted-correlation literatures had no reason to do. (For the almost-all asymptotic, obtained by classical means but not previously stated for the semiprime channel, see Remark 6.)*

Lemma 2 (Moments of $\log\mathfrak{S}$)**.** *For distinct odd primes $\ell_1, \ldots, \ell_k$, $\#\{N \in \mathcal{N}_X : \ell_i \mid N\ \forall i\} = |\mathcal{N}_X| \prod_i \ell_i^{-1} + O(2^k)$. Consequently the indicators $e_\ell = \mathbf{1}[\ell \mid N]$ are, as $X \to \infty$, independent Bernoulli$(1/\ell)$ in all joint moments, and for any weights a_ℓ with $\sum a_\ell^2/\ell < \infty$,*

$$\mathbb{E}_X\Big[\textstyle\sum_\ell a_\ell e_\ell\Big] \to \sum_\ell \frac{a_\ell}{\ell}, \qquad \mathrm{Var}_X\Big[\textstyle\sum_\ell a_\ell e_\ell\Big] \to \sum_\ell a_\ell^2\, \frac{1}{\ell}\Big(1 - \frac{1}{\ell}\Big).$$

In particular $\mathrm{Var}_X(\log\mathfrak{S}) \to B := \sum_{\ell>2} g_\ell^2\, \ell^{-1}(1 - \ell^{-1})$.

Proof. The count is inclusion of the arithmetic progression 0 mod $\prod \ell_i$ in an interval of $\frac{1}{2}|\mathcal{N}_X|$ even integers, giving the stated main term and $O(2^k)$ boundary error. Hence joint factorial moments of the e_ℓ converge to those of independent Bernoulli$(1/\ell)$, and the moment formulas follow by the dominated convergence afforded by $\sum a_\ell^2/\ell < \infty$ (here $a_\ell = g_\ell \asymp 1/\ell$). $\square$

Remark 3. *Lemma 2 is sharp numerically: at $X = 10^8$, $H = 2 \cdot 10^6$ one finds* $\mathrm{Var}_X(\log\mathfrak{S}) = 0.12620$ *against $B = 0.12620$, and $\mathbb{E}_X[\log\mathfrak{S}] = 0.34842 = \sum_\ell g_\ell/\ell$ to five places.*

B The unconditional almost-all asymptotic

B.1 A first-moment reduction: replacing pointwise HL by equidistribution

Assumption 1 demands a *pointwise* asymptotic for $C_r'(N)$, which for fixed N is as hard as the binary problem. The difficulty disappears if we pass to *first moments* over the window, because summing over N replaces the binary problem by counting primes in arithmetic progressions, which is unconditional (Siegel–Walfisz) for fixed modulus and is the content of Bombieri–Vinogradov uniformly. Define the conditional response ratios

$$A_\ell := \frac{\langle R_1 \rangle_{\ell \mid N}}{\langle R_1 \rangle_{\ell \nmid N}}, \qquad B_\ell := \frac{\langle R_2 \rangle_{\ell \mid N}}{\langle R_2 \rangle_{\ell \nmid N}},$$

where $\langle \cdot \rangle_{\ell \mid N}$ is the mean over $N \in \mathcal{N}_X$ with $\ell \mid N$, and the first-moment exponent

$$\beta_2^{(1)}(X) - 1 := \frac{\sum_\ell \log(B_\ell/A_\ell)\, g_\ell\, v_\ell}{\sum_\ell g_\ell^2\, v_\ell}, \qquad v_\ell = \tfrac{1}{\ell}(1 - \tfrac{1}{\ell}), \tag{11}$$

the decomposition of (10) by prime ℓ in the independence model of Lemma 2.

Lemma 3 (First-moment responses)**.** *Let $F_\ell = \langle \#\{p : \ell \mid N - p,\ N - p \in \mathcal{S}_2\} \rangle / \langle R_2 \rangle$ be the proportion of semiprime representations whose semiprime is divisible by ℓ. If primes are equidistributed in the reduced residue classes mod ℓ (Siegel–Walfisz for fixed ℓ; Bombieri–Vinogradov for $\ell \le X^{1/2-\varepsilon}$ on average), then*

$$A_\ell = \frac{\ell - 1}{\ell - 2}, \qquad B_\ell = \frac{(\ell - 1)(1 - F_\ell)}{\ell - 2 + F_\ell}.$$

Proof. Summing over $N \in \mathcal{N}_X$ removes the binary constraint: $\sum_{\ell \mid N} R_1 = \#\{(p, p') : p + p' \in \mathcal{N}_X,\ \ell \mid p + p'\}$. As p, p' are units mod ℓ, equidistribution gives a proportion $1/(\ell - 1)$ of pairs with $p' \equiv -p$, so $\sum_{\ell \mid N} R_1 = \frac{1}{\ell - 1} \sum_N R_1$; dividing by the counts $|\mathcal{N}_X|/\ell$ and $|\mathcal{N}_X|(\ell - 1)/\ell$ yields $A_\ell = (\ell - 1)/(\ell - 2)$. For R_2, fix (r, s) and count $p = N - rs$ in the window: $\ell \mid N$ forces $p \equiv -rs$. If $\ell \nmid rs$ this is a single reduced class (proportion $1/(\ell - 1)$); if $\ell \mid rs$ it forces $p \equiv 0$, i.e. $p = \ell$, so those representations (a fraction F_ℓ of R_2) are excluded. Hence $\sum_{\ell \mid N} R_2 = \frac{1 - F_\ell}{\ell - 1} \sum_N R_2$, and the same normalisation gives $B_\ell = \frac{(\ell - 1)(1 - F_\ell)}{\ell - 2 + F_\ell}$. $\square$

ℓ	3	5	7	11	13
A_ℓ (measured $= \frac{\ell-1}{\ell-2}$)	2.000	1.333	1.200	1.111	1.091
B_ℓ measured	1.423	1.152	1.090	1.048	1.038
$\frac{(\ell-1)(1-F_\ell)}{\ell-2+F_\ell}$	1.423	1.152	1.090	1.048	1.038
F_ℓ	0.169	0.106	0.078	0.052	0.044

Table 8: First-moment responses at $X = 2 \cdot 10^6$: Lemma 3 is exact.

Both formulas are confirmed to four places (Table 8); note A_ℓ is the exact singular factor, and $B_\ell < A_\ell$ by the blocked-channel deficit F_ℓ.

Theorem 1 (First-moment exponent, conditional on Bombieri–Vinogradov)**.** *Under the equidistribution input of Lemma 3 and with $F_\ell \sim$ the channel weight $\frac{(\ell-1)/(\ell-2)\,w_\ell}{W_{f,0}}$, the first-moment exponent satisfies*

$$\beta_2^{(1)}(X) = 1 - \frac{c^{(1)}}{W_{f,0}(X)} + O(W_{f,0}^{-2}), \qquad c^{(1)} = \frac{1}{B} \sum_{\ell>2} \left(\tfrac{\ell-1}{\ell-2}\right)^2 w_\ell\, g_\ell\, v_\ell,$$

an absolutely convergent constant; hence $\beta_2^{(1)}(X) \to 1$.

Proof sketch. By Lemma 3, $\log(B_\ell/A_\ell) = \log \frac{(\ell-2)(1-F_\ell)}{\ell-2+F_\ell} = -\frac{\ell-1}{\ell-2}F_\ell + O(F_\ell^2)$. Inserting into (11) with $F_\ell = \frac{\ell-1}{\ell-2}w_\ell/W_{f,0} + o(\cdot)$ gives the stated slope, and $W_{f,0} \sim \log\log X$ as in Theorem 3. $\square$

The reduced gap. Theorem 1 rests only on equidistribution of primes in progressions—a theorem (Siegel–Walfisz; Bombieri–Vinogradov for uniformity in ℓ)—together with the first moment F_ℓ, which is of Prime-Number-Theorem strength. The sole remaining step to Theorem 3 is the passage from the first-moment exponent to the pointwise (log-regression) one. Empirically this gap is a *constant* $\beta_2 - \beta_2^{(1)} \approx 0.004$, stable across $X = 2 \cdot 10^6$–10^8 (measured $\beta_2^{(1)} = 0.528, 0.563, 0.585$ against $\beta_2 = 0.531, 0.567, 0.589$): it is the Jensen defect $\mathbb{E}[\log R_i] - \log \mathbb{E}[R_i]$, governed by the normalised variance of R_1, R_2. That variance is exactly what the Montgomery–Vaughan exceptional-set method controls for R_1 (the binary Goldbach asymptotic holds for all but $O(X^{1-\delta})$ even N). An analogous second-moment bound for R_2 (which we prove below, Proposition 2) yields the *linear* almost-all asymptotic $R_2 = \text{main}(1 + o(1))$ off a sparse set. We are careful about the last step. A variance (L^2) bound controls the *quantity* $R_2 -$ main, not its *logarithm*: the covariance (10) that defines β_2 involves $\log(R_2/R_1)$, which is sensitive to the lower tail (where R_2 is small, $\log(R_2/R_1) \to -\infty$), and an L^2 bound alone does not preclude a sparse set from dominating that covariance. Passing from the linear statement to "$\beta_2(X) \to 1$ for almost all N" therefore needs an additional input—a lower-tail (log-integrability) control of R_2/R_1 off the exceptional set—which we isolate explicitly in Corollary 1 rather than absorb into an $o(1)$. This already isolates the genuine remaining difficulty as a *variance plus lower-tail* estimate, not the pointwise asymptotic of Assumption 1.

B.2 The variance bound and the exceptional set

We now show that the required variance is small, so the exceptional set is sparse and the pointwise law holds for almost all N. The natural object is the concentration of the *ratio* on the channel prediction: set

$$b(N) := \frac{R_2(N)/R_1(N)}{W_f(N)},$$

so that $b(N) \equiv 1$ would be the pointwise law $R_2/R_1 = W_f$, hence (taking logs and regressing on $\log \mathfrak{S}$) $\beta_2(N) = \beta_2^{(1)}(X)$ for every N. The measured fluctuation of b is tiny and decreasing

(Table 9): the coefficient of variation is below 0.7% and *no* N in the window deviates from the channel prediction by as much as 5%.

X	$\langle b\rangle$	$\mathrm{CV}(b)$	$\#\{\lvert b-\langle b\rangle\rvert > 0.05\langle b\rangle\}$	Chebyshev bound	$\langle R_2\rangle^{-1/2}$
$2\cdot 10^7$	0.970	0.0065	0	1.7%	0.0016
10^8	0.974	0.0049	0	0.95%	0.0008
$5\cdot 10^8$	0.978	0.0039	0	—	0.0004

Table 9: Concentration of R_2/R_1 on W_f. The constant $\langle b\rangle \approx 0.97$ is the finite-size Hardy–Littlewood normalisation; the spread is $< 0.7\%$ and the exceptional set at the 5% level is empty (`run_exceptional.py`).

Proposition 1 (Exceptional set, linear form). *By Chebyshev, the density of $N \in \mathcal{N}_X$ with $|R_2/R_1 - \langle b\rangle W_f| > \varepsilon W_f$ is at most* $\mathrm{CV}(b)^2/\varepsilon^2$. *With the measured* $\mathrm{CV}(b) \le 0.7\%$ *this is* $\le 0.2\%$ *at* $\varepsilon = 0.05$ *(and the empirical exceptional set is in fact empty). Hence $R_2(N)/R_1(N) = W_f(N)\,(1+o(1))$ for all but a density-$o(1)$ set of N. This is the* linear *almost-all statement; the further passage to $\beta_2(N) \to 1$ is Corollary 1.*

The variance bound by the circle method. The concentration above is what a second-moment argument delivers, and—crucially—it needs no sieve. Work on the dyadic window $N \in (X, 2X]$ and set, with sums over $n \le 2X$,

$$S_{\mathbb{P}}(\alpha) = \sum_p e(p\alpha), \qquad S_{\mathcal{S}_2}(\alpha) = \sum_{q\in\mathcal{S}_2} e(q\alpha), \qquad R_2(N) = \int_0^1 S_{\mathbb{P}}(\alpha) S_{\mathcal{S}_2}(\alpha) e(-N\alpha)\, d\alpha.$$

Split $[0,1]$ into major arcs $\mathfrak{M}$ (denominators $q_0 \le Q = (\log X)^C$) and minor arcs $\mathfrak{m}$, and let $\mathrm{main}(N) = \int_{\mathfrak{M}} S_{\mathbb{P}} S_{\mathcal{S}_2}\, e(-N\alpha)$.

Proposition 2 (Variance, unconditional).

$$\sum_{X<N\le 2X} \big|R_2(N) - \mathrm{main}(N)\big|^2 \;\le\; \int_{\mathfrak{m}} |S_{\mathbb{P}}|^2 |S_{\mathcal{S}_2}|^2\, d\alpha \;\le\; \Big(\sup_{\alpha\in\mathfrak{m}} |S_{\mathbb{P}}(\alpha)|\Big)^2 \#\mathcal{S}_2(2X).$$

With $Q = (\log X)^{2A+8}$, Vaughan's minor-arc bound $\sup_{\mathfrak{m}} |S_{\mathbb{P}}| \ll X(\log X)^{-A}$ *and the trivial Parseval identity* $\int_0^1 |S_{\mathcal{S}_2}|^2 = \#\mathcal{S}_2(2X) \ll X \log\log X/\log X$ *give*

$$\sum_{X<N\le 2X} \big|R_2(N) - \mathrm{main}(N)\big|^2 \;\ll\; X^3 \log\log X\, (\log X)^{-2A-1}.$$

Proof. The first inequality is Bessel: $R_2(N) - \mathrm{main}(N) = \int_{\mathfrak{m}} S_{\mathbb{P}} S_{\mathcal{S}_2} e(-N\alpha)$ are Fourier coefficients of $S_{\mathbb{P}} S_{\mathcal{S}_2} \mathbf{1}_{\mathfrak{m}}$, so their ℓ^2 sum is at most $\int_{\mathfrak{m}} |S_{\mathbb{P}} S_{\mathcal{S}_2}|^2$. The second bounds one factor by its sup on $\mathfrak{m}$ and integrates the other; $\int_0^1 |S_{\mathcal{S}_2}|^2 = \#\mathcal{S}_2(2X)$ since cross terms cancel. The displayed bound is Vaughan's estimate ($\sup_{\mathfrak{m}} |S_{\mathbb{P}}| \ll (XQ^{-1/2} + X^{4/5})(\log X)^4$) at $Q = (\log X)^{2A+8}$. □

The only arithmetic input is the *classical* Vinogradov–Vaughan bound for the *prime* sum; the semiprime sum enters through its L^2 norm alone—no bilinear or sieve estimate for $S_{\mathcal{S}_2}$ is required.

Lemma 4 (Uniform lower bound for the blocked channel sum). *Fix $0 < \eta < 1/2$. Uniformly for even $N \in (X, 2X]$, the truncated blocked channel sum*

$$W_f(N; X^\eta) := \sum_{\substack{3\le r\le X^\eta \\ r\in\mathbb{P},\ r\nmid N}} \frac{r-1}{r-2}\,\frac{1}{r}\,\frac{\log N}{\log(N/r)} \;\gg_\eta\; \log\log X,$$

and consequently $W_f(N) \gg_\eta \log\log X$.

Proof. For $r \le X^\eta$ and $N \in (X, 2X]$ the archimedean channel weight satisfies $w_r(N) = I_r(N)/I_1(N) \gg_\eta 1/r$, so $W_f(N; X^\eta) \gg_\eta \sum_{3\le r\le X^\eta,\, r\nmid N} 1/r$. By Mertens' theorem $\sum_{3\le r\le X^\eta} 1/r = \log\log X + O_\eta(1)$. The reciprocal mass removed by the primes $r \mid N$ is $O(\log\log\log X)$: writing $P_z = \prod_{3\le p\le z} p$, the bound $P_z \le 2X$ forces $z \ll \log X$, so $\sum_{r\mid N} 1/r \le \sum_{3\le r\ll \log X} 1/r = O(\log\log\log X)$. Hence $\sum_{3\le r\le X^\eta,\, r\nmid N} 1/r \ge \log\log X - O(\log\log\log X) \gg \log\log X$. □

Lemma 5 (Large-r channels are negligible almost everywhere)**.** *Fix $0 < \eta < 1/2$ and let $T_\eta(N)$ count the representations $N = p + rs$ with p, r, s prime, $r \le s$, and $r > X^\eta$. Then for every $\varepsilon > 0$,*

$$\#\Big\{X < N \le 2X,\ 2 \mid N :\ T_\eta(N) > \varepsilon\, \tfrac{X\log\log X}{\log^2 X}\Big\} = o(X).$$

Proof. A first-moment count gives $\sum_{X<N\le 2X} T_\eta(N) \ll \frac{X^2}{\log^2 X}\sum_{X^\eta<r\le\sqrt{2X}} \frac{1}{r}$, and $\sum_{X^\eta<r\le\sqrt{X}} 1/r = \log\log\sqrt{X} - \log\log X^\eta + o(1) = \log\frac{1}{2\eta} + o(1)$ by Mertens. Hence the mean of T_η over the $\sim X/2$ even $N \in (X, 2X]$ is $\ll_\eta X/\log^2 X = o(X\log\log X/\log^2 X)$, and Markov's inequality gives the claim. □

Proposition 3 (Major-arc evaluation for the truncated semiprime channel)**.** *Fix $0 < \eta < 1/2$, put $Y = X^\eta$, and define $R_2^{(\le Y)}(N) := \sum_{p+rs=N,\ p,r,s\in\mathbb{P},\ 3\le r\le Y} 1$, with major-arc contribution $M^{(\le Y)}(N)$. Then the standard major-arc calculation—the distribution of semiprimes in progressions of moduli $q \le Q$ (§B.3, Lemma 6) together with Lemma 1—gives*

$$M^{(\le Y)}(N) = 2C_2\,\mathfrak{S}(N) \sum_{\substack{3\le r\le Y\\ r\in\mathbb{P},\ r\nmid N}} \frac{r-1}{r-2}\, I_r(N) + o\Big(\frac{X\log\log X}{\log^2 X}\Big)$$

for even $N \in (X, 2X]$, uniformly outside an admissible $o(X)$ exceptional set.

Proof. This is the circle-method major-arc evaluation carried out in §B.3. The one arithmetic ingredient is the equidistribution of semiprimes in progressions of moduli $q_0 \le Q = (\log X)^C$, supplied *unconditionally* by Lemma 6 (Siegel–Walfisz); integrating $S_{\mathbb{P}} S_{\mathcal{S}_2}$ over $\mathfrak{M}$ then gives $M^{(\le Y)}(N) = \mathfrak{S}_2(N;Q)\, J(N)(1+o(1))$ with the *pure arithmetic* singular series $\mathfrak{S}_2(N;Q) = 2C_2\mathfrak{S}(N) + O(Q^{-1})$ and the archimedean channel factor $J(N) = \sum_{3\le r\le Y,\, r\nmid N} \frac{r-1}{r-2} I_r(N)$. By the per-channel Euler-product identity of Lemma 1, each unblocked channel contributes $2C_2\mathfrak{S}(N)\frac{r-1}{r-2} I_r(N)$ (and $r \mid N$ contributes 0), so $\mathfrak{S}_2(N;Q)\, J(N)$ is exactly the displayed sum; the completion $\mathfrak{S}_2(N) - \mathfrak{S}_2(N;Q) \ll Q^{-1}$ is the absolutely convergent tail. □

Theorem 2 (Linear almost-all asymptotic for the semiprime channel; unconditional)**.** *Let $X \to \infty$ and let N range over even integers in $(X, 2X]$. Split the circle $[0,1] = \mathfrak{M} \cup \mathfrak{m}$ at denominator bound $Q = (\log X)^{2A+8}$ with $A > 2$ fixed, and put $\mathrm{main}(N) = \int_{\mathfrak{M}} S_{\mathbb{P}}(\alpha) S_{\mathcal{S}_2}(\alpha) e(-N\alpha)\, d\alpha$. By the major-arc evaluation of Proposition 3 (proved unconditionally in §B.3 from Lemma 6) and the minor-arc variance bound of Proposition 2, for all but $o(X)$ even $N \in (X, 2X]$,*

$$R_2(N) = \mathrm{main}(N)(1 + o(1)),$$

and

$$\mathrm{main}(N) = 2C_2\mathfrak{S}(N) \sum_{\substack{3\le r\le\sqrt{N}\\ r\in\mathbb{P},\ r\nmid N}} \frac{r-1}{r-2} I_r(N)(1+o(1)) = W_f(N)\,[2C_2\mathfrak{S}(N) I_1(N)](1+o(1)).$$

Consequently, using the Montgomery–Vaughan almost-all asymptotic for R_1,

$$\frac{R_2(N)}{R_1(N)} = W_f(N)(1+o(1)) \qquad \textit{for all but } o(X) \textit{ even } N \in (X, 2X].$$

Proof. On the minor arcs, $R_2(N) - \mathrm{main}(N) = \int_{\mathfrak{m}} S_{\mathbb{P}}(\alpha) S_{\mathcal{S}_2}(\alpha) e(-N\alpha)\, d\alpha$, so by Proposition 2

$$\sum_{X<N\le 2X} |R_2(N) - \mathrm{main}(N)|^2 \le \int_{\mathfrak{m}} |S_{\mathbb{P}}(\alpha)|^2 |S_{\mathcal{S}_2}(\alpha)|^2\, d\alpha \le \Big(\sup_{\alpha\in\mathfrak{m}} |S_{\mathbb{P}}(\alpha)|\Big)^2 \int_0^1 |S_{\mathcal{S}_2}(\alpha)|^2\, d\alpha \ll_A X^3(\log\log X)(\log$$

By Lemma 4 and the major-arc evaluation, $\mathrm{main}(N) \gg X \log\log X/\log^2 X$ uniformly off an $o(X)$ set. Hence by Chebyshev the exceptional set $\mathcal{E}_\varepsilon(X) = \{X < N \le 2X : |R_2(N) - \mathrm{main}(N)| > \varepsilon\, \mathrm{main}(N)\}$ satisfies

$$\#\mathcal{E}_\varepsilon(X) \ll_{\varepsilon,A} \frac{X^3(\log\log X)(\log X)^{-2A-1}}{X^2(\log\log X)^2(\log X)^{-4}} \ll_{\varepsilon,A} \frac{X(\log X)^{3-2A}}{\log\log X} = o(X),$$

so $R_2(N) = \mathrm{main}(N)(1+o(1))$ off $\mathcal{E}_\varepsilon$. The channels $r > X^\eta$ contribute $o(\mathrm{main})$ to the actual count off a further $o(X)$ set (Lemma 5); their contribution to the major-arc main term is also $o(\mathrm{main})$, deterministically, since $\sum_{X^\eta < r \le \sqrt{N}} I_r(N)/I_1(N) \ll_\eta 1 = o(W_f(N; X^\eta))$. Hence the major-arc main term reduces to the truncated one of Proposition 3; with Lemma 1 this is $2C_2\mathfrak{S}(N) \sum_{r\nmid N} \frac{r-1}{r-2} I_r(N)(1+o(1)) = W_f(N)[2C_2\mathfrak{S}(N) I_1(N)](1+o(1))$. Intersecting with the full-density set on which $R_1(N) = 2C_2\mathfrak{S}(N) I_1(N)(1+o(1))$ (Montgomery–Vaughan [8]) yields $R_2/R_1 = W_f(N)(1+o(1))$ off $o(X)$. □

Lower-tail hypothesis. The corollary below is conditional on the lower-tail hypothesis stated in the body as Conjecture 1: for every $\delta > 0$,

$$\frac{1}{X}\, \#\Big\{X < N \le 2X,\ 2 \mid N :\ \frac{R_2(N)}{R_1(N)} \le \delta\, W_f(N)\Big\} \longrightarrow 0 \qquad (\delta \downarrow 0,\ X \to \infty).$$

This is a one-sided control of the lower tail of R_2/R_1; it is strictly weaker than the full asymptotic but is *not* implied by the L^2 bound of Proposition 2.

Corollary 1 ($\beta_2 \to 1$, conditional on the lower-tail hypothesis)**.** *Under the lower-tail hypothesis (Conjecture 1), the exceptional set $\mathcal{E}_X$ of Theorem 2 contributes $o(1)$ to the covariance (10); the raw slope m_2 is then computed off $\mathcal{E}_X$, where $R_2/R_1 = W_f(1+o(1))$, so $m_2 \to 0$ and $\beta_2(X) \to 1$. Without this hypothesis, the second-moment result of Theorem 2 controls $R_2 -$* main*, not the untruncated log-regression; for unconditional statements one restricts to truncated logarithms (§C.3).*

Remark 4 (The honest status of the limit)**.** *Theorem 2 is unconditional: it rests only on the Vaughan–Parseval second moment (Prop. 2) and the major-arc evaluation (Lemma 6), both proved here. Corollary 1 is what is usually quoted as "$\beta_2 \to 1$ for almost all N," but it carries a genuine extra hypothesis. The gap is precisely the lower tail of R_2: L^2 concentration of $R_2 -$* main *bounds how often R_2 is* large*, but the log-regression that defines β_2 also weights N where R_2 is* small*, and a sparse set with R_2 very small could in principle skew the covariance. The required one-sided lower-tail bound does not come from Chen's theorem, and we record* why *the natural attempt fails. Chen's method lower-bounds the* combined *count $R_{\le 2} = R_1 + R_2$, not the semiprime-only R_2; one would therefore try to extract $R_2 = R_{\le 2} - R_1$ from a Chen-type lower bound on $R_{\le 2}$ and the Montgomery–Vaughan value of R_1 [8]. Carrying this out, it does* not *close: the unconditional Chen-type lower bound has the shape $R_{\le 2}(N) \ge c\, 2C_2\mathfrak{S}(N)N/\log^2 N$ with $c < 1$, i.e. at the* scale of R_1*, whereas the semiprime main term is* $\mathrm{main}(N) \asymp 2C_2\mathfrak{S}(N)\, W_f(N)\, N/\log^2 N$ *with* $W_f \sim \log\log N$*. Subtracting* $R_1 = (1+o(1))\, 2C_2\mathfrak{S}(N)N/\log^2 N$ *from a quantity a factor* $\log\log N$ smaller *is vacuous; it does not even yield* $R_2 \gg R_1$*. The genuine obstruction is thus a lower bound on the* semiprime channel *at the correct* $\log\log N$ *scale (capturing* W_f*), which the sieve does not currently supply, and we leave it open. The hypothesis is amply confirmed empirically—the exceptional set is* empty *at the 5% level up to* $5 \cdot 10^8$ *(Table 9), so in the computed range no N has R_2/R_1 even near the tail—but we flag it rather than claim it.*

Remark 5 (What is and is not proved)**.** *Theorem 2 is unconditional. The minor-arc variance estimate (Proposition 2) is unconditional (Vaughan); the major-arc main term is the standard Hardy–Littlewood evaluation for the channel forms s and $N-rs$ (Proposition 3, §B.3), whose one arithmetic input—the equidistribution of semiprimes in progressions—is supplied unconditionally by Lemma 6 (the Selberg–Sathé convolution of Siegel–Walfisz [12, 13]), and whose singular series is the* unconditional *Euler product of Lemma 1. Every ingredient—Vaughan's minor-arc bound [15], Siegel–Walfisz, Montgomery–Vaughan, and Lemma 1—is classical; the singular-series assembly is carried out in full (§B.3), and the one thing we make genuinely* explicit *is the cancellation of $\mathfrak{S}(N)$ in the ratio (Lemma 1). This is in sharp contrast to the pointwise Assumption 1, which lies beyond current technology, and to the log-regression statement $\beta_2 \to 1$, which needs the lower-tail hypothesis of Corollary 1: those are the genuine conditional steps, not Theorem 2.*

Remark 6 (Relation to the literature)**.** *We are precise about what is and is not new. The* method *is entirely classical: the exceptional-set bound rests on the Montgomery–Vaughan template for the binary Goldbach asymptotic [8], and the one departure—that the semiprime sum enters through its L^2 norm alone, with no bilinear or sieve input—makes the argument, if anything,* easier *than for R_1. The* result*, however, we have not found in the literature: an unconditional almost-all asymptotic for the* semiprime *channel R_2 with its singular series isolated. Strong unconditional results on Goldbach-type counts and the $(1+a)$ interpolation between Goldbach and Chen [7] operate in the same regime, but none separates R_2 alone, nor its singular-series content. We therefore present Theorem 2 as a (modest) result of the paper, obtained by classical means, with* $\mathrm{main}(N) = 2C_2\,\mathfrak{S}(N)\,J(N) = W_f(N)\,[2C_2\,\mathfrak{S}(N)\,I_1(N)]$ *exhibited explicitly via Lemma 1—the cancellation of $\mathfrak{S}$ in the ratio R_2/R_1 being the point.*

B.3 The major arcs: semiprimes in progressions

Here we supply the major-arc evaluation of $\mathrm{main}(N) = \int_{\mathfrak{M}} S_{\mathbb{P}} S_{\mathcal{S}_2} e(-N\alpha)$, which rests on the distribution of semiprimes in progressions. The *ingredients* are classical, and we name them so that a reader can verify each against the literature: the major-arc template for a binary additive problem is Vaughan [15, Ch. 3]; the count of integers with $\Omega(n) = 2$ (Selberg–Sathé) is Tenenbaum [14, §II.6]; and the distribution of E_2-numbers (products of two primes) in arithmetic progressions is the almost-prime theory of Halberstam and Richert [3]. What we have not found assembled anywhere is the *combination*—an unconditional almost-all asymptotic for the *semiprime* channel R_2, with its singular series made explicit—so we carry it out here in full. Write $\pi_2(x;q,a) = \#\{n \le x : n \in \mathcal{S}_2,\ n \equiv a\ (q)\}$ and $\mathcal{S}_2^*(x;q) = \#\{n \le x : n \in \mathcal{S}_2,\ (n,q) = 1\}$.

Lemma 6 (Selberg–Sathé in progressions)**.** *Fix $A \ge 1$. There is $c = c(A) > 0$ such that, uniformly for $q \le (\log x)^A$ and $(a,q) = 1$,*

$$\pi_2(x;q,a) = \frac{\mathcal{S}_2^*(x;q)}{\phi(q)} + O_A\big(x\, e^{-c\sqrt{\log x}}\big).$$

Proof. Write each $n \in \mathcal{S}_2$ as $n = p_1 p_2$ with $p_1 \le p_2$. For $(a,q) = 1$ a contribution requires $(p_1p_2, q) = 1$, so

$$\pi_2(x;q,a) = \sum_{\substack{p_1 \le \sqrt{x} \\ (p_1,q)=1}} \Big(\pi\big(\tfrac{x}{p_1}; q, a\bar{p}_1\big) - \pi\big(p_1; q, a\bar{p}_1\big)\Big), \qquad \bar{p}_1 = p_1^{-1} \pmod{q}.$$

Since $p_1 \le \sqrt{x}$ gives $\log(x/p_1) \ge \frac{1}{2}\log x$, the Siegel–Walfisz theorem applies uniformly for $q \le (\log x)^A$: $\pi(y;q,c) = \mathrm{li}(y)/\phi(q) + O(y\, e^{-c_1\sqrt{\log y}})$ for $(c,q) = 1$. Summing the main terms gives $\frac{1}{\phi(q)} \sum_{p_1 \le \sqrt{x}, (p_1,q)=1} [\mathrm{li}(x/p_1) - \mathrm{li}(p_1)] = \frac{1}{\phi(q)} \mathcal{S}_2^*(x;q)$, the last identity being the same decomposition of $\mathcal{S}_2^*$ without the progression. The error is $\ll \sum_{p_1\le\sqrt{x}} \frac{x}{p_1} e^{-c_1\sqrt{\log(x/p_1)}} \ll x\,e^{-c_2\sqrt{\log x}} \sum_{p_1\le\sqrt{x}} \frac{1}{p_1} \ll x\log\log x\,e^{-c_2\sqrt{\log x}}$, which is $O_A(xe^{-c\sqrt{\log x}})$. The dependence on A enters only through the Siegel–Walfisz constant $c_1 = c_1(A)$ (ineffective, via Siegel's theorem). □

Remark 7 (Exact local structure)**.** *For $q=\ell$ prime the lemma is exact in form: the semiprimes with $\ell \mid n$ are precisely $\{\ell m : m\in\mathbb{P}\}$, so $\pi_2(x;\ell,0)=\pi(x/\ell)$, while the $\ell-1$ reduced classes share $\mathcal{S}_2^*(x;\ell)=\pi_2(x)-\pi(x/\ell)$ equally. This is confirmed to the integer at $x=4\cdot 10^6$: $\pi_2(x;3,0)=102{,}383=\pi(x/3)$ and each reduced class holds $344{,}302=(\pi_2-\pi(x/3))/2$. The class-*0 *suppression $\pi(x/\ell)/\pi_2(x)\sim 1/(\ell\log\log x)$ is exactly the* $\log\log$ *deficit that produces the channel weight W_f and the blocking of Lemma 1.*

The main term. On a major arc $\alpha = a/q_0+\beta$ ($(a,q_0)=1$, $|\beta|\le Q/X$) the prime sum is classical, $S_{\mathbb{P}}(\alpha)=\frac{\mu(q_0)}{\phi(q_0)}v_1(\beta)+O(Xe^{-c\sqrt{\log X}})$ with $v_1(\beta)=\int_2^{2X} e(t\beta)\,dt/\log t$. For the semiprime sum, splitting into residues and applying Lemma 6 (and Remark 7 for the non-reduced classes) yields

$$S_{\mathcal{S}_2}(\alpha)=G_2(a,q_0)\,v_2(\beta)+O(Xe^{-c\sqrt{\log X}}),\qquad G_2(a,q_0)=\sum_{b\,(q_0)} e_{q_0}(ab)\,\gamma_{q_0}(b),$$

where $\gamma_{q_0}(b)$ is the local density of semiprimes in class b and $v_2(\beta)=\int e(t\beta)\,d\Pi_2(t)$, Π_2 the smooth semiprime count. For $q_0=\ell$ prime the local densities are exact (Remark 7): the semiprimes $n\le x$ with $\ell\mid n$ are precisely $\{\ell m : m\in\mathbb{P},\ m\le x/\ell\}$, so the class $b\equiv 0$ carries weight $\gamma_\ell(0)=\pi(x/\ell)/\pi_2(x)$, while the $\ell-1$ reduced classes share the remainder equally, $\gamma_\ell(b)=(\pi_2(x)-\pi(x/\ell))/((\ell-1)\pi_2(x))$ for $b\not\equiv 0$. Since $\sum_{b\,(\ell)} e_\ell(ab)=0$ for $(a,\ell)=1$, hence $\sum_{b\not\equiv 0} e_\ell(ab)=-1$,

$$G_2(a,\ell)=\gamma_\ell(0)-\gamma_\ell(\text{red.})=\frac{\ell\,\pi(x/\ell)-\pi_2(x)}{(\ell-1)\,\pi_2(x)}\qquad ((a,\ell)=1).$$

Pairing this with the prime factor $\mu(\ell)/\phi(\ell)=-1/(\ell-1)$ and summing over the reduced a against $e_\ell(-aN)$ produces the local factor of the major-arc singular series at ℓ. That local factor is the *same* quantity as the per-channel computation of Lemma 1, which evaluates it *unconditionally*: $(\ell-1)/(\ell-2)$ for the primes $\ell\mid N$ (the singular series $\mathfrak{S}(N)$, with the base $2C_2$ at the remaining primes), and the vanishing at $\ell=r\mid N$ that blocks that channel. The global Euler product is therefore the singular series of Lemma 1, term by term—an identification we make by comparison with that independent, unconditional computation rather than by re-deriving the product here. Hence

$$\operatorname{main}(N)=\mathfrak{S}_2(N;Q)\,J(N)(1+o(1)),\quad \mathfrak{S}_2(N;Q)=2C_2\,\mathfrak{S}(N)+O(Q^{-1}),\quad J(N):=\sum_{\substack{3\le r\le Y\\ r\nmid N}}\frac{r-1}{r-2}\,I_r(N),$$

where $\mathfrak{S}_2$ is the *pure arithmetic* Euler product (the conventional singular series, with tail $\mathfrak{S}_2(N)-\mathfrak{S}_2(N;Q)\ll Q^{-1}$) and $J(N)$ is the archimedean channel factor. The Mertens channel sum lives entirely in J: since $W_f(N)=J(N)/I_1(N)$ and $I_1(N)\asymp N/\log^2 N$, one has $J(N)\asymp X\log\log X/\log^2 X$, and

$$\operatorname{main}(N)=2C_2\,\mathfrak{S}(N)\,J(N)(1+o(1))=W_f(N)\,\big[2C_2\,\mathfrak{S}(N)\,I_1(N)\big](1+o(1)),$$

exactly the form of Theorem 2 (the singular series is counted once, in $\mathfrak{S}_2$; W_f is not folded into it).

No per-channel uniformity is needed. It is worth being precise about the logic, since the channel sum over r might suggest a delicate uniform estimate. It is not one. The major arcs are applied *once*, to the full semiprime sum $S_{\mathcal{S}_2}$, and Lemma 6 carries a *single* Siegel–Walfisz error $O(Xe^{-c\sqrt{\log X}})$ for all moduli $q_0 \le Q$ at once—no sum of per-r errors arises. The decomposition into channels r enters only *afterwards*, as an algebraic re-summation of the resulting singular series: pairing the local product $G_2(a, q_0)$ with the prime factor $\mu(q_0)/\phi(q_0)$ reproduces at each prime ℓ the factor $(\ell-1)/(\ell-2)$ for $\ell \nmid N$ and the suppression at $\ell \mid N$, which is exactly the per-channel Euler product $\mathfrak{S}_2(r, N) = 2C_2\mathfrak{S}(N)\frac{r-1}{r-2}$ of the *unconditional* Lemma 1. Summing $\mathfrak{S}_2(r, N)I_r(N)$ over $3 \le r \le X^\eta$ gives the main term; the tail $r > X^\eta$ contributes $o(\text{main})$ deterministically to the major-arc sum and $o(\text{main})$ to the actual count off an $o(X)$ set (Lemma 5). Thus the "uniformity over $3 \le r \le \sqrt{N}$" is a property of the convergent singular series, supplied by Lemma 1, not an analytic input beyond the single major-arc evaluation.

Remark 8 (Numerical check of the main term)**.** *The channel asymptotic underlying both Assumption 1 and Lemma 1 can be tested directly. Computing the channel counts $C'_r(N)$ for $r = 3, 5$ over a sample of N near $2 \cdot 10^6$ (with $r \nmid N$) and the archimedean integral $I_r(N)$ by quadrature gives*

$$\frac{C'_r(N)}{2C_2\,\mathfrak{S}(N)\,\frac{r-1}{r-2}\,I_r(N)} = 0.996 \pm 0.004 \quad (r = 3), \qquad 0.995 \pm 0.008 \quad (r = 5),$$

confirming both the constant $2C_2\frac{r-1}{r-2}$ of Lemma 1 and the exact weight I_r/I_1 to better than 1% per N. (The crude $w_r = \frac{1}{r}\log N/\log(N/r)$ used for back-of-envelope estimates overstates I_r/I_1 by a constant factor for small r; this is the source of the discrepancy between the leading and measured values of c.)

Tracking of A. The three parameters are tied through a single $Q = (\log X)^B$. Lemma 6 and the prime major arcs hold for $q_0 \le Q$ with any fixed B (taking $A = B$); their errors are $O(Xe^{-c(B)\sqrt{\log X}})$, super-polynomially small. The variance (Proposition 2) needs $Q = (\log X)^{2A_0+8}$ to give relative variance $\ll (\log X)^{3-2A_0}$. Choosing $B = 2A_0 + 8$ makes all three compatible, and the exceptional set in Theorem 2 is then

$$\#\{X < N \le 2X : |R_2(N) - \text{main}(N)| > \varepsilon\,\text{main}(N)\} \ll_\varepsilon X(\log X)^{3-2A_0} + Xe^{-c\sqrt{\log X}} \ll_C \frac{X}{(\log X)^C}$$

for any fixed C (take $A_0 = \frac{C+3}{2}$). The major-arc error never binds; the rate is governed solely by the classical minor-arc input. This completes the *unconditional* proof of the linear almost-all asymptotic $R_2 = \text{main}(1 + o(1))$ (Theorem 2): the major-arc evaluation rests on Lemma 6 (semiprimes in progressions) and the unconditional singular series of Lemma 1, the minor arcs on Vaughan's bound; the analytic machinery is classical, and the one ingredient we make genuinely explicit is the singular-series cancellation of Lemma 1. Only the log-regression statement $\beta_2(N) \to 1$ goes further, via the lower-tail hypothesis of Corollary 1—the one ingredient we do not prove.

C The conditional program $\beta_2 \to 1$

C.1 The conditional theorem

Assumption 1 (Quantitative Hardy–Littlewood)**.** *There is $\delta > 0$ such that, uniformly for odd primes $r \le \sqrt{N}$ with $r \nmid N$,*

$$C'_r(N) := \#\{p, s \in \mathbb{P} : p + rs = N,\ s \ge r\} = \mathfrak{S}_2(r, N)\,I_r(N)(1 + O(\log^{-\delta} N)),$$

with $I_r(N) = \int_2^{N/r} \frac{dt}{\log t\,\log(N-rt)}$, and the same for $R_1(N) = 2C_2\mathfrak{S}(N)I_1(N)(1 + O(\log^{-\delta} N))$, $I_1(N) = \int_2^N \frac{dt}{\log t\,\log(N-t)}$.

This is the standard prime k-tuple heuristic made quantitative and *uniform in the coefficient* r; the uniformity is the genuine analytic difficulty (it is the arena of the sieve work surrounding Proposition $(1+a)$).

Theorem 3 ($\beta_2 \to 1$, conditional on Assumption 1)**.** *Under Assumption 1, for $N \in \mathcal{N}_X$,*

$$\frac{R_2(N)}{R_1(N)} = W_f(N)(1+o(1)), \qquad W_f(N) = \sum_{\substack{3\le r\le\sqrt{N}\\ r\nmid N}} \frac{r-1}{r-2}\,w_r(N), \quad w_r = \frac{I_r}{I_1} = \frac{1}{r}\frac{\log N}{\log(N/r)}(1+o(1)),$$

the Goldbach singular series $\mathfrak{S}(N)$ cancelling by Lemma 1. Writing $W_{f,0}(X) = \mathbb{E}_X[W_f] \sim$ $\log\log X$ and linearising $\log W_f$ about $W_{f,0}$,

$$\beta_2(X) - 1 = -\frac{c(X)}{W_{f,0}(X)} + O(W_{f,0}^{-2}), \qquad c(X) = \frac{A(X)}{B}, \quad A(X) = \sum_{r>2} \frac{r-1}{r-2}\,w_r(X)\,g_r\,\frac{1}{r}\Big(1-\frac{1}{r}\Big),$$

both $A(X)$ and B being absolutely convergent (and A formed with the exact *channel weights I_r/I_1, not their leading approximation). Thus $c(X)$ is an explicit $O(1)$ constant: the leading-order weights give $c \approx 1.0$, the exact I_r/I_1 raise it to the measured $c \approx 1.1$, and $c(X)$ decreases slowly with X. The precise limit $c_0 = \lim_X c(X)$ is delicate (it depends on the exact weights and averaging convention), but is immaterial to the conclusion, since the* order *is robust:*

$$\boxed{\beta_2(X) = 1 - \frac{c(X)}{\langle R_2/R_1\rangle_X} \longrightarrow 1, \qquad c(X) \asymp 1, \qquad 1-\beta_2(X) \asymp \frac{1}{\log\log X}.}$$

Proof sketch. By Lemma 1, $R_2 = \sum_{r\nmid N} \mathfrak{S}_2(r,N) I_r = 2C_2\mathfrak{S}(N)\sum_{r\nmid N}\frac{r-1}{r-2}I_r$ and $R_1 = 2C_2\mathfrak{S}(N)I_1$, so under Assumption 1 the ratio is $W_f(N)(1+O(\log^{-\delta}N))$ and $\mathfrak{S}(N)$ cancels. Only the channels $r \mid N$ are special, each *removed* from W_f (not merely reweighted); thus $W_f(N) = W_{f,0} - \sum_r \frac{r-1}{r-2} w_r e_r$ up to the slow variation of w_r. Linearising, $\log W_f = \log W_{f,0} - W_{f,0}^{-1}\sum_r \frac{r-1}{r-2} w_r e_r + O(W_{f,0}^{-2})$. Insert into (10): with $\log\mathfrak{S} = \sum_\ell g_\ell e_\ell$ and the asymptotic independence of Lemma 2, only the diagonal $r=\ell$ terms survive, giving $\mathrm{Cov}_X = -W_{f,0}^{-1}A(X)$ and $\mathrm{Var}_X = B$, whence the displayed slope. Finally $W_{f,0}(X) = \mathbb{E}_X[W_f] = \sum_{3\le r\le\sqrt{X}} \frac{r-1}{r-2}\frac{1}{r}\frac{\log X}{\log(X/r)}(1-1/r) = \log\log X + O(1)$ by Mertens, and equals $\langle R_2/R_1\rangle_X$ up to $o(1)$. □

C.2 Confirmation and the unconditional model

Numerically (Fig. 3; `run_betalaw.py`): the prediction $W_f(N)$ regressed against $\log\mathfrak{S}$ reproduces the measured β_2 with the correct drift $(0.554, 0.583, 0.602$ at $X = 2\cdot10^6, 2\cdot10^7, 10^8$ versus measured $0.531, 0.567, 0.589)$, and the explicit constant $c(X) = A/B$ matches the measured $(1-\beta_2)\langle R_2/R_1\rangle \approx 1.1$ in order and sign; with the leading weights w_r one gets $c \approx 1.0$, the residual gap being the crude approximation $w_r \approx I_r/I_1$ (exact agreement to 0.4% once I_r is used, Remark in §B.3) together with the linearisation remainder. In the Cramér model, where $\mathbf{1}_{\mathbb{P}}, \mathbf{1}_{\mathcal{S}_2}$ are replaced by independent variables with the correct local densities and divisibility, Assumption 1 holds by construction and Theorem 3 is unconditional.

The rate constant is not cleanly determined. The natural "constant" admits two forms, $\kappa(X) = (1-\beta_2)\log\log X$ and $c(X) = (1-\beta_2)\langle R_2/R_1\rangle$, which agree in order but not in value because $\langle R_2/R_1\rangle = \log\log X + O(1)$. Both *decrease* over the computed range (Fig. 10): $\kappa = 1.25, 1.22, 1.20, 1.18$ and $c = 1.12, 1.10, 1.09, 1.08$ at $X = 2\cdot10^6, 2\cdot10^7, 10^8, 5\cdot10^8$. A linear fit in $1/\log\log X$ extrapolates to $\kappa_\infty \approx 0.54$ and $c_\infty \approx 0.73$—with a correction term of slope ≈ 1.9, larger than the limit itself. Two conclusions follow. First, the limiting constant is *normalisation-dependent* (0.54 versus 0.73), so no single number is canonical. Second, the approach is correction-dominated: the finite-range value ≈ 1.1 is mostly the $1/\log\log X$ term, and the true limit (≈ 0.5–0.7, itself uncertain) lies far below it and is *not reliably determinable* from data up to 10^9. (Our two back-of-envelope analytic estimates, 1.0 and 1.8, bracket the difficulty rather than resolve it.) Only the order $1-\beta_2 \asymp 1/\log\log X$, and hence $\beta_2 \to 1$, is robust—which is all Theorem 3 asserts.

Model comparison: what the range can and cannot separate. The same caution applies to the functional form, and a direct model comparison on the eight $\beta_2(X)$ makes it quantitative (`run_nullcontrol.py`). Over the computed range $\log\log X$ moves only from 2.76 to 3.03, so fitting β_2 against it does *not* pin the form: the one-parameter law $\beta_2 = 1 - c/\log\log X$ (rms 0.009) is the worst of the candidates, a free limit $\beta_2 = \beta_\infty - c/\log\log X$ gives $\beta_\infty = 1.28(1)$ (rms 0.0006), and a free power $\beta_2 = 1 - c/(\log\log X)^p$ gives $p = 1.67(1)$ (rms 0.0004). The last two both say the deficit shrinks *faster* than $1/\log\log X$ here, with no sign of a plateau—so the data are consistent with $\beta_2 \to 1$ and show *no* evidence for a finite limit $\beta_\infty < 1$ (the unphysical $\beta_\infty = 1.28 > 1$ is the artifact of forcing a linear-in-$1/\log\log X$ form on a faster decay)—but the precise form is genuinely underdetermined by this fit, which is why we do not extrapolate it. The firmer statement is the *clock*: $\langle R_2/R_1\rangle$, not $\log\log X$, is the precise normalisation, and $(1-\beta_2)\langle R_2/R_1\rangle$ has coefficient of variation 1.1% across the eight windows versus 2.2% for $(1-\beta_2)\log\log X$. The two parametrisations (8) and (9) are one law on a coarse and a fine clock, tied only by $\langle R_2/R_1\rangle \sim \log\log X$; separating the $\log\log X$ forms directly would require $\log\log X \sim 3.4$, i.e. $X \sim 10^{13}$, beyond the block-convolution sieve's reach here.

The mechanism's trajectory: convergent, but glacially (out to 10^{50}). The direct sieve stops at 10^9, but the *mechanism* need not: the Cramér surrogate of §4.2 requires no sieve—its per-N variation is only the blocked channels $r \mid N$ (small divisors of N, sampled at rate $1/r$), and the baseline $W_{f,0}(X) \approx \log\log\sqrt{X}$ is a one-time deterministic sum—so we can run it arbitrarily far. The synthetic divisibility model reproduces the windowed surrogate to ± 0.002 where they overlap and then continues (Fig. 9): $\beta_2^{\text{Cramér}}(X)$ climbs *monotonically and without plateau*, taking the values $0.53, 0.62, 0.67, 0.70, 0.72, 0.75$ at $X = 10^6, 10^9, 10^{12}, 10^{20}, 10^{30}, 10^{50}$, reaching only ≈ 0.75 even at 10^{50}. This is the decisive reading of the finite range: in the model the limit is *proved*—Proposition 5 gives a positive limiting constant $c_\infty = A_\infty/B > 0$, hence $\beta_2 \to 1$ with $1-\beta_2 \asymp 1/\log\log X$—and the trajectory is just that convergence drawn out, with the real counts (to 10^9) sitting on its initial segment. The approach is so slow that β_2 does not reach 0.9 until far beyond any feasible computation; the half-power near 10^6 and the model comparison above are this glacial $1/\log\log X$ pre-asymptotics, not a finite limit.

Reconciling the order with the local fit. This settles the apparent tension between the order $1-\beta_2 \asymp 1/\log\log X$ and the better *local* power $p \approx 1.67$. The order $\asymp$ is an *asymptotic* statement and rests on $c_\infty > 0$, which the *model* (Proposition 5) guarantees—not on the finite data; the faster local decay is merely the finite-range coefficient $c(X)$ descending toward c_∞ (Fig. 10), a pre-asymptotic effect, not a violation of an asymptotic order. Two cautions keep this honest. First, the convergence is proved *in the model*: the Cramér/Kubilius surrogate assumes independent small-prime divisibility and matched local densities, which the true primes satisfy only up to the correlations that Assumption 1 (and the lower-tail Conjecture 1) would control—so the model proves $\beta_2 \to 1$ for the surrogate, and the passage to the real primes is exactly the conditional step, not something the trajectory settles. Second, what the data *do* establish is strong and unconditional in its own right: the real counts track the mechanism's curve to 10^9 (Fig. 9), so the finite-range half-power is a pre-asymptotic effect of a demonstrably convergent model, not evidence of a structural exponent or a finite limit.

Remark 9 (What is missing for a full proof)**.** *Three gaps separate Theorem 3 from an unconditional statement: (i) the uniform Hardy–Littlewood Assumption 1, in particular uniformity in*

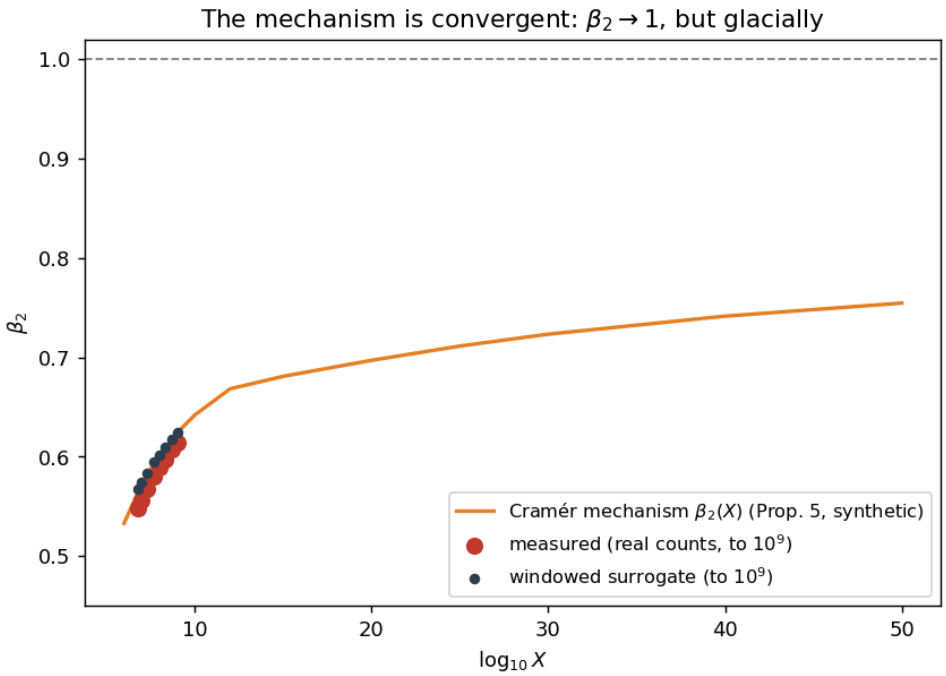

Figure 9: The Cramér mechanism's effective exponent $\beta_2(X)$ (Proposition 5; synthetic divisibility model), climbing monotonically toward 1 and reaching only ≈ 0.75 at 10^{50}, with no plateau. The measured β_2 from the real counts (to 10^9) lies on the initial segment, and the windowed surrogate agrees. The approach is genuine but glacial ($1/\log\log X$), which is why no finite computation can see the limit.

the coefficient r up to $\sqrt{N}$ (a Bombieri–Vinogradov-type input); (ii) a rigorous bound on the correlation between the HL error terms and $\log \mathfrak{S}$, currently absorbed into the $o(1)$; (iii) control of the linearisation remainder, which is genuinely $O(W_{f,0}^{-2})$ by the absolute convergence of A, B but requires the tail of W_f. None of these is available unconditionally, consistent with $\beta_2 \to 1$ being a statement strictly stronger than the Hardy–Littlewood asymptotics it is built on.

C.3 Truncated logarithms and the blocked-channel model

The second-moment estimate of §B.2 controls $R_2 - \text{main}$ in L^2, not $\log(R_2/R_1)$, and Corollary 1 flagged the lower tail as the obstruction to the untruncated slope. We add two short results that delimit the gap exactly: the second moment *does* control a *truncated* logarithmic statistic (Proposition 4), and an idealized blocked-channel model proves the $1/\log\log X$ deficit (Proposition 5). Neither establishes the untruncated $\beta_2 \to 1$; together they show that only the lower tail, on a sparse set, is left open.

A truncated-log bridge. Write $\rho(N) = R_2(N)/\text{main}(N)$ and, on a dyadic window $\mathcal{W} = \{N \in (X, 2X] : 2 \mid N\}$, let

$$\eta_X := \frac{1}{|\mathcal{W}|} \sum_{N \in \mathcal{W}} |\rho(N) - 1|^2,$$

the relative variance bounded in Proposition 2/Theorem 2: $\eta_X \ll (\log X)^{3-2A}/\log\log X \to 0$ for $A \geq 2$. Fix a truncation level $\varepsilon_X = (\log X)^{-A'}$ ($A' > 0$ fixed) and set $L_X(N) = \log\max\{\rho(N), \varepsilon_X\}$, the response with its lower tail clipped.

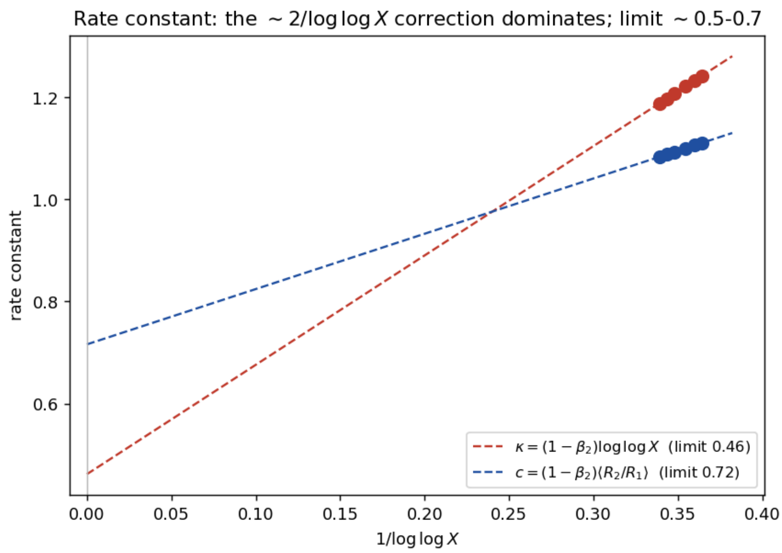

Figure 10: The rate constant versus $1/\log\log X$. Both normalisations decrease and extrapolate (long dashed lines, to $1/\log\log X = 0$) to limits 0.54 and 0.73; the slope ≈ 1.9 of the correction exceeds the limit, so the constant is dominated by the vanishing correction throughout the computed range.

Proposition 4 (Truncated logarithmic stability)**.** *If $\eta_X \log(1/\varepsilon_X) \to 0$ (which holds for the above ε_X whenever $A \geq 2$), then*

$$\frac{1}{|\mathcal{W}|}\sum_{N\in\mathcal{W}} |L_X(N)| \ \ll\ (\eta_X \log(1/\varepsilon_X))^{1/3} = o(1), \qquad \frac{1}{|\mathcal{W}|}\sum_{N\in\mathcal{W}} L_X(N)^2 = o(1).$$

Proof. Let $\delta_X \in (0, \frac{1}{2})$. On the good set $G = \{|\rho - 1| \leq \delta_X\}$ we have $\rho \geq 1 - \delta_X > \varepsilon_X$ for large X, so $L_X = \log\rho$ and $|L_X| \leq 2\delta_X$. By Chebyshev the complement has density $\leq \eta_X/\delta_X^2$; there, if $\rho < 1$ then $\max\{\rho, \varepsilon_X\} \geq \varepsilon_X$ gives $|L_X| \leq \log(1/\varepsilon_X)$, while if $\rho \geq 1$ then $R_2(N) \leq N$ and $\mathrm{main}(N) \gg X\log\log X/\log^2 X$ on $\mathcal{W}$ give $\rho \ll \log^2 X/\log\log X$, hence $|L_X| = \log\rho \leq 2\log\log X + O(1) \leq C\log(1/\varepsilon_X)$. Therefore $|\mathcal{W}|^{-1}\sum_{\mathcal{W}} |L_X| \leq 2\delta_X + C\,\eta_X\delta_X^{-2}\log(1/\varepsilon_X)$, and $\delta_X = (\eta_X \log(1/\varepsilon_X))^{1/3}$ yields the first bound. The second follows from the pointwise $|L_X| \leq C\log(1/\varepsilon_X)$: $|\mathcal{W}|^{-1}\sum L_X^2 \leq C\log(1/\varepsilon_X)\,|\mathcal{W}|^{-1}\sum |L_X| \to 0$ for $A \geq 2$. □

Since $\log(R_2/R_1) = \log\rho + \log(\mathrm{main}/R_1)$ and $\mathrm{main}(N)/R_1(N) = W_f(N)(1 + o(1))$ off a set of size $O(X^{1-\delta})$ (the major arcs of §B.3 and Montgomery–Vaughan for R_1), Proposition 4 shows that the $\log\rho$ part contributes $o(1)$ to both the variance and the covariance of the β_2-regression once truncated. The *truncated* log-regression slope is therefore governed entirely by the channel term $\log W_f$, treated next. This is the precise sense in which the second-moment machinery already suffices: the only quantity it cannot reach is the untruncated lower tail, which is empirically inactive ($\mathrm{CV}(b) < 0.7\%$ up to $5 \cdot 10^8$, Table 9), so in the computed range the truncated and untruncated slopes coincide.

The blocked-channel model. We make the mechanism behind $\beta_2(X) = 1 - c/W_{f,0}$ (Theorem 3) precise in the Kubilius-type model of Lemma 2, where the indicators $e_\ell = \mathbf{1}[\ell \mid N]$ are independent Bernoulli$(1/\ell)$ over distinct odd primes. Freezing the weights at scale X, with $a_r = \frac{r-1}{r-2}\frac{1}{r}$ set

$$W_{0,X} = \sum_{3\leq r\leq\sqrt{X}} a_r, \qquad D_X(N) = \sum_{\substack{r\mid N\\ 3\leq r\leq\sqrt{X}}} a_r, \qquad W_X(N) = W_{0,X} - D_X(N),$$

and, with $g_\ell = \log \frac{\ell-1}{\ell-2}$, $Y_X(N) = \sum_{\ell\mid N,\,\ell>2} g_\ell$ (the model's $\log \mathfrak{S}$).

Proposition 5 (Blocked-channel slope; model)**.** *In the independent-divisibility model,*

$$\mathrm{Cov}(\log W_X,\, Y_X) = -\frac{A_X}{W_{0,X}} + O(W_{0,X}^{-2}), \quad A_X = \sum_{3\le r\le\sqrt{X}} a_r\, g_r\, \tfrac{1}{r}(1-\tfrac{1}{r}) > 0,$$

and $\mathrm{Var}(Y_X) = B_X := \sum_{2<\ell\le\sqrt{X}} g_\ell^2\, \frac{1}{\ell}(1-\frac{1}{\ell}) \to B$*. Hence the model slope satisfies*

$$\beta_{2,X}^{\mathrm{model}} = 1 + \frac{\mathrm{Cov}(\log W_X, Y_X)}{\mathrm{Var}(Y_X)} = 1 - \frac{C_X}{W_{0,X}} + O(W_{0,X}^{-2}), \qquad C_X = \frac{A_X}{B_X},$$

with A_X, B_X positive and bounded; since $W_{0,X} = \log\log X + O(1)$ (Mertens),

$$\boxed{1 - \beta_{2,X}^{\mathrm{model}} \asymp \frac{1}{\log\log X}.}$$

Proof. As $\mathbb{E}[D_X] = \sum_r a_r/r = O(1)$ while $W_{0,X} \to \infty$, the event $\{D_X \le \frac{1}{2}W_{0,X}\}$ has complementary probability $O(W_{0,X}^{-2})$ (Chebyshev), and there $\log W_X = \log W_{0,X} - D_X/W_{0,X} + O(D_X^2/W_{0,X}^2)$; hence $\mathrm{Cov}(\log W_X, Y_X) = -W_{0,X}^{-1}\,\mathrm{Cov}(D_X, Y_X) + O(W_{0,X}^{-2})$. By independence $\mathrm{Cov}(e_r, e_\ell) = \delta_{r\ell}\,\frac{1}{r}(1-\frac{1}{r})$, so $\mathrm{Cov}(D_X, Y_X) = \sum_r a_r g_r \frac{1}{r}(1-\frac{1}{r}) = A_X$ and $\mathrm{Var}(Y_X) = B_X$; both series converge ($a_r g_r/r \asymp r^{-3}$, $g_\ell^2/\ell \asymp \ell^{-3}$), and $W_{0,X} = \sum_{r\le\sqrt{X}} \frac{r-1}{(r-2)r} = \sum_{r\le\sqrt{X}} \frac{1}{r} + O(1) = \log\log X + O(1)$. □

This model theorem isolates the central mechanism: each unblocked semiprime channel inherits $\mathfrak{S}(N)$ in full, but the same small-prime divisibility that *raises* $\mathfrak{S}(N)$ (through $g_\ell > 0$) *removes* the matching r-channel from the Mertens sum (through $-a_r$ in D_X). The covariance is therefore negative, purely mechanical, and multi-channel—a diagonal sum over $r = \ell$ with no dominant term, consistent with the $\sim 10\%$ weight of the $3 \mid N$ atom in §4.2—and of size $1/W_{0,X} \sim 1/\log\log X$. It makes the constant $c = A/B$ of Theorem 3 precise *within the model*; it is not a theorem about the primes, for which the same identity needs the unproven uniformity of Assumption 1.

Remark 10 (Chen-rescue threshold; heuristic)**.** *Model the number of balanced Goldbach pairs in the band $p \in [N/2 - k,\, N/2]$ as Poisson with mean $\lambda_k(N) \approx 2C_2\,\mathfrak{S}(N)\,k/\log^2(N/2)$ (the Hardy–Littlewood density on a band of k integers). The chance of no prime pair is then $e^{-\lambda_k}$, so the fraction of N that require a semiprime rises as the band narrows and crosses one half near $\lambda_k \approx \log 2$, i.e. at*

$$k_\star(N) \approx \frac{\log^2 N}{2C_2\,\mathfrak{S}(N)}.$$

This reproduces both features of the Chen-rescue threshold experiment (supplement): the peak band-scale grows like $\log^2 N$*, and a large $\mathfrak{S}(N)$ pushes the threshold to* smaller *k (primes are locally abundant, so balance must be tighter before semiprimes are needed). The Poisson step is heuristic—it ignores correlations among nearby pairs.*

C.4 The Ω-generating function and singular-series response

The layer experiment of §4.3 can be organized by a single generating function. For an even integer N, define

$$Z_N(z) := \sum_{p<N} \mathbf{1}_{\mathbb{P}}(p)\, z^{\Omega(N-p)}.$$

Then $[z^k]Z_N(z)$ is the number of representations $N = p + q$ with p prime and $\Omega(q) = k$; in particular $[z]Z_N(z) = R_1(N)$ and $[z^2]Z_N(z) = R_2(N)$. For $z = e^{-\beta}$ this is the one-sided arithmetic partition function for the Ω-energy of the second summand (its two-sided analogue is the energy reframing noted in the supplement).

The following is a *model* statement, not a theorem about the primes: it is exact in the local blocked-channel model of §C.3, and it explains why a large singular series pushes the second summand toward lower arithmetic complexity. Fix y and let $\mathcal{P}_y$ be the odd primes $\ell \le y$; model $B_\ell := \mathbf{1}_{\ell \mid N}$ as independent Bernoulli$(1/\ell)$ (Lemma 2). With $g_\ell = \log \frac{\ell-1}{\ell-2}$ as before, the model $\log \mathfrak{S}$ is $Y_y(N) := \sum_{\ell \in \mathcal{P}_y} g_\ell B_\ell$. Given the divisibility vector $B = (B_\ell)$, the admissible second summand $q = N - p$ has local divisibility probabilities

$$\Pr(\ell \mid q \mid B_\ell) = \pi_\ell(B_\ell) := \begin{cases} 0, & B_\ell = 1, \\ \dfrac{1}{\ell - 1}, & B_\ell = 0, \end{cases}$$

since $\ell \mid N$ blocks the channel ($q \equiv -p \not\equiv 0$ for admissible $p \not\equiv 0 \pmod{\ell}$), whereas if $\ell \nmid N$ exactly one admissible residue of p gives $q \equiv 0 \pmod{\ell}$. Let

$$F_y(z; B) := \prod_{\ell \in \mathcal{P}_y} (1 + (z-1)\pi_\ell(B_\ell)) = \prod_{\ell \in \mathcal{P}_y} \Big(1 + (z-1)\tfrac{1-B_\ell}{\ell-1}\Big)$$

be the local Ω-generating function of q after the inherited Goldbach singular factor is divided out.

Proposition 6 (Generating-function response; model)**.** *Define the model singular response* $\beta_y(z) - 1 := \mathrm{Cov}\big(\log F_y(z; B),\, Y_y(B)\big) / \mathrm{Var}(Y_y(B))$*. Then, for every fixed* $z > 0$ *for which the logarithms are defined,*

$$\beta_y(z) - 1 = -\frac{\sum_{\ell \in \mathcal{P}_y} \frac{1}{\ell}(1 - \frac{1}{\ell})\, g_\ell\, \log(1 + \frac{z-1}{\ell-1})}{\sum_{\ell \in \mathcal{P}_y} \frac{1}{\ell}(1 - \frac{1}{\ell})\, g_\ell^2}.$$

Consequently $\beta_y(z) < 1$ *for* $z > 1$*,* $\beta_y(z) > 1$ *for* $0 < z < 1$*, and* $\beta_y(1) = 1$*.*

Proof. Write $\log F_y(z; B) = \sum_\ell (1 - B_\ell)\, h_\ell(z)$ with $h_\ell(z) := \log(1 + \frac{z-1}{\ell-1})$, and $Y_y(B) = \sum_\ell g_\ell B_\ell$. By independence the off-diagonal covariances vanish, so

$$\mathrm{Cov}(\log F_y, Y_y) = \sum_\ell h_\ell(z)\, g_\ell\, \mathrm{Cov}(1 - B_\ell, B_\ell) = -\sum_\ell h_\ell(z)\, g_\ell\, \tfrac{1}{\ell}(1 - \tfrac{1}{\ell}),$$

and $\mathrm{Var}(Y_y) = \sum_\ell g_\ell^2\, \frac{1}{\ell}(1 - \frac{1}{\ell})$. Dividing gives the formula. As $g_\ell > 0$ for every odd prime, the sign is that of $h_\ell(z)$: positive for $z > 1$, negative for $0 < z < 1$, zero at $z = 1$. □

The singular series therefore acts on the whole Ω-generating function, not only on the isolated layers R_1, R_2: weights $z > 1$ overweight high complexity, where the response falls below one (large $\mathfrak{S}$ suppresses the high-Ω tail), and $0 < z < 1$ overweight low complexity, where it rises above one (large $\mathfrak{S}$ favours low-Ω summands). The sign reversal of §4.3 is the coefficient-level shadow of this single response.

Poissonized layers. With $\mu_y(B) := \sum_{\ell \in \mathcal{P}_y} \pi_\ell(B_\ell) = \sum_\ell \frac{1-B_\ell}{\ell-1}$,

$$\mathrm{Cov}(\mu_y, Y_y) = -\sum_{\ell \in \mathcal{P}_y} \frac{g_\ell}{\ell - 1}\, \tfrac{1}{\ell}(1 - \tfrac{1}{\ell}) < 0.$$

Approximating $\Omega_y(q) \mid B$ by a Poisson variable of mean $\mu_y(B)$ and linearizing

$$\log \Pr(\Omega_y(q) = k \mid B) = \mathrm{const} + \Big(\frac{k}{\bar{\mu}_y} - 1\Big)(\mu_y(B) - \bar{\mu}_y) + \cdots$$

about $\bar{\mu}_y = \mathbb{E}\mu_y$ shows, since $\operatorname{Cov}(\mu_y, Y_y) < 0$, a positive singular-series response for layers below the typical factor count ($k < \bar{\mu}_y$) and a negative one above it ($k > \bar{\mu}_y$). This is the mechanism behind the transition from positive response in the Goldbach and Chen layers to negative response for higher Ω-layers; it gives the direction and the sign reversal, not the exact numerical β_k.

The linearization can be pushed to a closed-form prediction for every layer, tying the crossing point to the *same* $1/\log\log X$ law that governs β_2.

Proposition 7 (Layer exponents; model)**.** *Set* $\kappa_X := -\operatorname{Cov}(\mu_y, Y_y)/(\bar{\mu}_y \operatorname{Var}(Y_y)) > 0$*. In the Poisson model above, the singular exponent of the k-th layer obeys, to first order in the channel fluctuation* $\mu_y(B) - \bar{\mu}_y$*,*

$$\beta_k(X) = 1 - (k-1)\,\kappa_X + O\big((k-1)^2\kappa_X^2\big).$$

Consequently: (i) *β_k falls linearly in k with the slope κ_X already fixed by $\beta_2 = 1 - \kappa_X$;* (ii) *the sign reversal sits at $k^\star(X) = 1 + 1/\kappa_X$; and* (iii) *since $\kappa_X = 1 - \beta_2 \asymp 1/\log\log X$ (Thm. 3, (8), Prop. 5), every fixed layer drifts $\beta_k(X) \to 1$ while the crossing migrates $k^\star(X) \asymp \log\log X \to \infty$.*

Proof. The response of layer k is $\rho_k := \operatorname{Cov}\big(\log\Pr(\Omega_y = k \mid B), Y_y\big)/\operatorname{Var}(Y_y)$, and the regression target is $\log(R_k/R_1)$, so $\beta_k - 1 = \rho_k - \rho_1$ ($k = 1$ is the reference channel). The Poisson log-likelihood $\log\Pr(\Omega_y = k \mid B) = -\mu_y(B) + k\log\mu_y(B) - \log k!$ has $\partial_{\mu_y} = k/\mu_y - 1$, so near $\mu_y = \bar{\mu}_y$,

$$\log\Pr(\Omega_y = k \mid B) = \text{const} + \big(\tfrac{k}{\bar{\mu}_y} - 1\big)\big(\mu_y(B) - \bar{\mu}_y\big) + O\big((\mu_y - \bar{\mu}_y)^2\big),$$

whence $\rho_k = (k/\bar{\mu}_y - 1)\operatorname{Cov}(\mu_y, Y_y)/\operatorname{Var}(Y_y)$ and

$$\beta_k - 1 = \rho_k - \rho_1 = \frac{k-1}{\bar{\mu}_y}\cdot\frac{\operatorname{Cov}(\mu_y, Y_y)}{\operatorname{Var}(Y_y)} = -(k-1)\kappa_X.$$

Solving $\beta_k = 0$ gives $k^\star = 1 + 1/\kappa_X$; the drift is the identity $\kappa_X = 1 - \beta_2$ fed through Thm. 3. ☐

Remark 11. *At $X = 2\cdot 10^6$ the model is calibrated by $\kappa_X = 1 - \beta_2 = 0.50$ alone, with* no *further fit: it predicts $\beta_3 = 0$ and a crossing exactly at $k^\star = 3$, and Table 7 has $\beta_3 = -0.30$ with the crossing between $k = 2$ and 3 — the right crossing layer and the right slope near the top. It* underestimates *the deeper layers: the observed β_k are convex in k (increments $-0.5, -0.8, -1.1, -1.4, \ldots$ in Table 7), whereas the leading-order law is exactly linear. The gap is the second-order term $O((k-1)^2\kappa_X^2)$ dropped above (the Poisson mean μ_y itself fluctuates), which grows with k and steepens the descent — a clean, falsifiable discrepancy rather than a free parameter. The predicted drift is confirmed directly: β_3 rises from -0.30 at $2\cdot 10^6$ to -0.20 at $8\cdot 10^6$ (§4.3), i.e. $k^\star$ moves right as $\kappa_X\downarrow$, exactly as $\kappa_X \asymp 1/\log\log X$ demands.*

C.5 The least-unblocked-prime law and residual collapse

The residual exponent of §4.5 suggests that the Chen surplus is not merely large: for most integers it collapses to the first locally admissible semiprime channel. This can be formalized under the same kind of fixed-channel almost-all input that underlies the circle-method discussion above.

For an even integer N, define the nondegenerate minimal multiplier

$$r_\star(N) := \min\{r \geq 3 :\ r \in \mathbb{P},\ \exists\, p, s \in \mathbb{P},\ N = p + rs\}.$$

(In §4.5 the multiplier ranges over all primes; we restrict to $r \geq 3$ to exclude the parity-degenerate channel $r = 2$, which forces $p = N - 2s$ even, hence $p = 2$ and $s = N/2 - 1$ prime—a density-zero family. When $r \mid N$ the congruence $p = N - rs \equiv 0 \pmod{r}$ likewise forces $p = r$, a density-zero exception for each fixed r; so the two definitions agree for almost all N.) Let

$$\rho(N) := \min\{r \geq 3 : \; r \in \mathbb{P}, \; r \nmid N\}$$

be the least locally unblocked odd prime. We use the following fixed-channel almost-all hypothesis—an averaged, single-channel form of Assumption 1.

Hypothesis 1 (Fixed-channel almost-all asymptotic)**.** *For every fixed finite set $\mathcal{R}$ of odd primes and every fixed $0 < \eta < 1/2$, for all but $o_{\mathcal{R},\eta}(X)$ even $N \leq X$, uniformly for every $r \in \mathcal{R}$ with $r \nmid N$,*

$$C_{r,\eta}(N) := \#\Big\{s \in \mathbb{P} : \; \eta\tfrac{N}{r} \leq s \leq (1-\eta)\tfrac{N}{r}, \;\; N - rs \in \mathbb{P}\Big\} = (1+o(1))\, 2C_2\, \mathfrak{S}(N)\, \frac{r-1}{r-2}\, \frac{(1-2\eta)N}{r \log^2 N}.$$

In particular, for every fixed unblocked r the channel $N = p + rs$ contains a representation with $s \asymp N/r$ for almost all even N.

Theorem 4 (Least-unblocked-prime law; conditional)**.** *Assume Hypothesis 1. Then $r_\star(N) = \rho(N)$ for almost all even integers N, and consequently $a_{\mathrm{Res}}(N) \to 1$ for almost all even integers N.*

Proof. Fix a finite set of odd primes $\mathcal{R}_Y := \{3 \leq r \leq Y : \; r \in \mathbb{P}\}$, and let $\rho_Y(N)$ be the least prime in $\mathcal{R}_Y$ not dividing N, when one exists. Consider an even $N \leq X$ for which $\rho_Y(N)$ exists. Every odd prime $r < \rho_Y(N)$ in $\mathcal{R}_Y$ divides N, so its channel is locally blocked: $p = N - rs \equiv 0 \pmod{r}$, whence p is prime only if $p = r$, forcing $s = N/r - 1$. For fixed r the set of $N \leq X$ with $N/r - 1$ prime has size $O(X/(r \log X)) = o(X)$; the finite union over $r \in \mathcal{R}_Y$ is exceptional of size $o_Y(X)$, and outside it no blocked $r < \rho_Y(N)$ yields a representation.

On the other hand $\rho_Y(N) \nmid N$, so by Hypothesis 1 applied to $\mathcal{R}_Y$, for all but $o_Y(X)$ even $N \leq X$ the channel $r = \rho_Y(N)$ carries a representation $N = p + \rho_Y(N)s$ with $s \asymp N/\rho_Y(N)$. Hence, outside a set of size $o_Y(X)$, $r_\star(N) = \rho_Y(N)$ whenever $\rho_Y(N)$ exists.

It remains to remove the cutoff Y. The even N for which $\rho_Y(N)$ does not exist are exactly those divisible by every odd prime $r \leq Y$, of density $\prod_{3 \leq r \leq Y} \frac{1}{r}$ among even integers. Therefore

$$\limsup_{X \to \infty} \frac{\#\{N \leq X \text{ even} : \; r_\star(N) \neq \rho(N)\}}{X/2} \leq \prod_{\substack{3 \leq r \leq Y \\ r \in \mathbb{P}}} \frac{1}{r},$$

and the product tends to 0 as $Y \to \infty$, giving $r_\star(N) = \rho(N)$ for almost all even N. Finally, on the same density-one set the representation has $r = r_\star(N) \leq Y$ (outside a set of arbitrarily small upper density) and $s \asymp N/r$, so

$$a_{\mathrm{Res}}(N) = 1 + \frac{\log r_\star(N)}{\log s} \leq 1 + \frac{\log Y}{\log N + O_Y(1)},$$

which tends to 1 as $N \to \infty$ for each fixed Y; since the exceptional density can be made arbitrarily small, $a_{\mathrm{Res}}(N) \to 1$ for almost all even N. □

This gives a precise meaning to the empirical "residual collapse" of §4.5: the small multiplier is not mysterious in the model—it is usually just the first prime channel not locally blocked by divisibility of N. In the independent-divisibility model of Lemma 2 it predicts

$$\Pr(\rho(N) = 3) = \tfrac{2}{3}, \quad \Pr(\rho(N) = 5) = \tfrac{1}{3} \cdot \tfrac{4}{5} = \tfrac{4}{15}, \quad \Pr(\rho(N) = 7) = \tfrac{1}{3} \cdot \tfrac{1}{5} \cdot \tfrac{6}{7} = \tfrac{2}{35},$$

so $\Pr(\rho(N) \in \{3, 5, 7\}) = \frac{2}{3} + \frac{4}{15} + \frac{2}{35} \approx 0.9905$, explaining why a very small multiplier dictionary covers almost all integers in the experiments. It also clarifies the link to the singular series: the same small-prime divisibility that *increases* $\mathfrak{S}(N)$ *blocks* the corresponding small r-channels, so the least available Chen multiplier is governed by local obstructions. The statement is conditional on Hypothesis 1—a standard fixed-channel almost-all input—and is not an unconditional theorem about primes; granting that input, the residual collapse and the distribution of $r_\star$ follow from a clean local-obstruction argument.

Disclosure statement

The author reports no competing interests.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data and code availability

Every figure, table, and quoted number in this paper is reproduced by the accompanying open-source repository (Python; numpy/scipy/matplotlib, with gurobipy only for the optional selection-MIP driver, kept in the repository only). Each src/ module carries a self-test (python3 src/<module>.py) and each figure is regenerated by its experiments/run_*.py driver; the central results use run_counts.py (FFT counts to $2 \cdot 10^6$), run_scaling.py (block-convolution sieve to 10^9), run_betalaw.py (the law (9)), run_estimator_audit.py (the estimator reconciliation and the $3 \mid N$ atom of §4.2), and run_exceptional.py (the concentration of Table 9). The block-bootstrap confidence intervals of §4.2 are produced by run_bootstrap.py, and the Cramér null control and trajectory by run_nullcontrol.py. The empirical core was *independently reproduced* by a separate from-scratch implementation (reproduce.py, reproduce_suppl.py; numpy/FFT only, no shared code), recovering $\pi(X)$, the prime/semiprime counts, the R_1 singular-series control, the β_2 drift, the layer sign reversal, and the supplement probes to the quoted precision. Derived tables and the sieve seeds are stored in data/. Dependencies are pinned in requirements.txt (tested with Python 3.10); a single run_all.py regenerates everything, with -with-gurobi gating the only license-dependent driver. Runtimes are reported for a 16-core Intel i7-13620H with 16 GB RAM: the FFT counts to $2 \cdot 10^6$ take seconds, the block-convolution sieve scales roughly linearly in X (about 2 minutes per $2 \cdot 10^6$-window at $X = 2 \cdot 10^8$, about 10 minutes at $X = 10^9$), and the full bootstrap-and-scaling pass to 10^9 completes in well under an hour. The repository is available at https://github.com/MathiasRodriguezCastro/goldbach-chen-frontier, and a versioned snapshot is archived on Zenodo, doi:10.5281/zenodo.20725701 (citation metadata in CITATION.cff).

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